Block 4
Financial Markets

UNIT 10 MONEY MARKETS

Structure

10.0 OBJECTIVES

Having completed this unit, you will be able to:

- explain the concept of the money market;
- spell out the composition of the money market;
- highlight the importance, role and functions of the money market in an economy;
- identify requisites of a good money market;
- describe the Indian money market;
- analyse the various regulatory mechanisms governing the working of the money market in India; and
- indicate the problems of the Indian money market.

10.1 INTRODUCTION

As we have learnt in previous block that financial institutions, financial markets and instruments together constitute the financial system. Thus, financial markets are one of the important constituents of financial system. The financial markets of all economies are broadly subdivided into two parts: (i) *Money Market*, and (ii) *Capital Market*. Both these markets provide avenues to surplus sectors such as households and institutions to deploy their funds and opportunity to corporate and government sectors to mobilize funds for their requirements. Financial markets across the world are organised differently to serve the need to save, borrow, invest, transact, and insure. They vary on regulations, ownership type, level of consolidation, etc.

The money market refers to a section of the financial market having financial instruments with high liquidity and short-term maturities that usually range from overnight to just under a year such as treasury bills and commercial papers.

The money market is considered a safe place to invest due to the high liquidity of securities. It has certain risks which investors should be aware of, one of them being default on securities such as commercial papers. The money market consists of various financial institutions and dealers, who make available to a broad range of borrowers and investors the opportunity to buy and sell various forms of short-term securities. The money market gives lesser returns to investors who invest in it but provides a variety of products. Withdrawing money from the money market is easier. The presence of an active and vibrant money market is an essential prerequisite for the growth and development of an economy. This unit will throw light on the various facets of the money market along with the structure and functions of the Indian money market.

The *money market* being an important segment of the financial market, not only reflects the impact of liquidity mismatch in the system but also operates as the first leg in transmitting monetary policy changes to the other parts of the financial system. In an integrated financial system, the occurrence of an event in one market of the financial system will have an impact on the other market of the system.

The money market *enables asset-liability management* in commercial banks. It provides an avenue for short-term investment in respect of cash held to maintain sufficient liquidity through its many *sub-markets*. The larger the number of sub-markets the broader and more developed the structure of the money market. The interest rates and the conditions prevailing in the money market influence the *rate of interest* and mobilization of resources in the capital market, the foreign exchange market and the gilt securities market. Thus, the development of these markets, to a large extent, is dependent on the development of the money market. The money market largely is a tool for the development of industry, infrastructure, agriculture, trading and commerce in any country.

10.2 CONCEPT AND FEATURES OF MONEY MARKET

10.2.1 Concept of Money Market

A money market is a market for short-term money and financial assets that are close substitutes for money. By the term 'Short-term' in the Indian context, it means a period of up to one year. *According to the Reserve Bank of India*, a money market is 'the centre for dealings, mainly of short-term character, in money assets. It meets the short-term requirements of borrowers and provides liquidity or cash to the lenders. It is the place where short-term surplus investible funds at the disposal of financial and other institutions and individuals are bid by borrowers' agents comprising institutions and individuals and also the government itself. The money market is essentially an over-the-phone market where transactions are conducted online or through telephonic communication with or without the help of brokers to be followed subsequently by written documents and exchange of instruments. It is a mechanism to clear short-term monetary transactions in an economy.

Individuals, institutions, and the government are economic agents that participate in the money market. These agencies create demand for money and also ensure the supply of money for a short-term period. The demand for money emanates from merchants, traders, brokers, manufacturers, speculators and even government institutions. The suppliers include commercial banks, insurance companies, non-banking financial concerns and the Central Bank of the country. The money market consists of many *sub-markets*. According to *Vaghul* Committee, "*The money market is a market for short-term financial assets that are close substitutes for money. The important feature of money market instruments is that it is liquid and can be turned over quickly at low cost, and it provides an avenue for equilibrating the short-term surplus funds of lenders and the requirements of borrowers*."

10.2.2 The Distinctive Features of the Money Market

i) The money market consists of negotiable instruments. It is used by many participants, including companies, to raise funds by selling commercial papers in the market. The money market is considered a safe place to invest due to the high liquidity of securities.

ii) The money market is a market for short-term financial assets which can be turned over quickly at a low cost. It provides an avenue for equilibrating the short-term surplus funds of lenders and the requirements of borrowers.

iii) It is one market but a collection of markets, such as, call money, notice money, repose, term money, treasury bills, commercial bills, certificate of deposits, commercial papers, inter-bank participation certificates, inter-corporate deposits, swaps futures, options, etc. and is concerned to deal in a particular type of assets. Its chief characteristic is its relative liquidity. All the sub-markets have close inter-relationships and free movement of funds from one sub-market to another. There has to be a network of a large number of participants which will add greater depth to the market.

iv) The activities in the money market tend to concentrate in some centre which serves a region or an area; the width of such area may vary considerably in some markets like London and New York which have become world financial centres. Where more than one market exists in a country, with screen-based trading and revolutions in information technology, such markets have rapidly become integrated into a national market. In India, Mumbai is emerging as a national market for money market instruments.

v) An important feature of the money market is its' impersonal character so that competition will be relatively pure. The money market is entirely different where no visible shops and counters exist.

vi) Trading is a wholesale process as the volume of funds or financial assets traded in the money market is very large.

10.3 OBJECTIVES AND FUNCTIONS OF MONEY MARKET

Objectives: The major objectives of money market are enlisted below:

i) A parking place to employ short-term surplus funds.

ii) Room for overcoming short-term deficits.

iii) Reasonable access to users of short-term funds to meet their requirements quickly, adequately and at reasonable costs.

iv) The Central Bank to influence and regulate liquidity in the economy through its intervention in this market.

The money market is an important part of the economy. It plays a very significant role in facilitating the various sectors of the economy in the following manner:

i) **Financing Trade:** The money market provides financing to local and international traders who are in urgent need of short-term funds. It provides a facility to discount bills of exchange, and thereby providing finance to pay for goods and services. International traders benefit from the acceptance houses and discount markets. The money market also makes funds available for other units of the economy such as agriculture and small-scale industries.

ii) **Help in Implementing Central Bank Policies**: The money market provides a base and mechanism for effective implementation of the monetary policy. Through the money market, the central bank can perform its policy-making function efficiently. Integrated money markets help the central bank to influence the sub-markets and thereby getting its monetary policy implemented. Central Bank ensures that the liquidity and the short-term interest rates are consistent with the monetary policy.

iii) **Growth of Business and Industries**: The money market is important for businesses because it allows companies with a temporary cash surplus to invest in short-term securities; conversely, companies with a temporary cash shortfall can sell securities or borrow funds on a short-term basis. In essence, the market acts as a repository for short-term funds and to finance the working capital needs. Although the money market does not provide long-term loans, it influences the capital market and can also help businesses obtain long-term financing. The capital market benchmarks its interest rates based on the prevailing interest rate in the money market.

iv) **Self-Sufficiency to Commercial Banks:** The money market provides commercial banks with a ready market where they can invest their excess reserves and earn interest while maintaining liquidity. Short-term investments such as bills of exchange can easily be converted to cash to support customer withdrawals. Also, when faced with liquidity problems, they can borrow from the money market on a short-term basis as an alternative to borrowing from the central bank. The advantage of this is that the money market may charge lower interest rates on short-term loans than the central bank typically does.

v) **Other functions**: Promotes economic growth by making funds available to various units in the economy such as agriculture, small-scale industries, etc. and makes available investment avenues for short-term periods. It helps in generating savings and investments in the economy. The money market provides non-inflationary sources of finance to the government. It is possible to raise short-term loans by issuing treasury bills which does not lead to an increase in the prices. In short, the functions of the money market are:

   i) It provides *an equilibrating mechanism* for evening out short-term surpluses and deficits.

   ii) A focal point for central bank intervention for influencing liquidity in the economy as per monetary policy.

   iii) Reasonable access to the users of short-term money to meet their requirements at realistic prices i.e., rate of interest.

10.4 REQUISITES FOR A GOOD FUNCTIONING MONEY MARKET

A good money market is one of the prerequisites *for a nation's economic progress* and the effective implementation of monetary policy. A money market is considered *'developed'* if it satisfies the following conditions:

i) Institutional development, relative political stability and a reasonably well-developed highly organised commercial banking and financial systems.

ii) The presence of a strong and efficient Central bank ensures credibility in the system, supervision of the players in the market and facilitation of the implementation of monetary policy by way of open market operations.

iii) Promote financial mobility in the form of inter-sectoral flows of funds.

iv) Continuous supply of negotiable securities such as bills of exchange, treasury bills, short-term Government bonds, etc. The market should have varied instruments with distinctive maturity and risk profiles to meet the varied appetites of the players in the market. Multiple instruments add strength and depth to the market.

v) Existence of several sub-markets, each specialising in a particular type of short-term asset. The larger the number of sub-markets, the broader and more developed will be the structure of the money market. The funds should move from one segment of the market to another to exploit the advantages of arbitrage opportunities. The sub-markets should form an integrated structure in which every segment of the money market is in an intimate relationship with each other. This is essential to ensure uniformity of interest rates in various sub-markets and also for the free flow of funds. This helps channelising savings into productive investments (like working capital).

vi) The market should be integrated with the rest of the markets in the financial system to ensure perfect equilibrium. Efficient payment systems for clearing and settlement of transactions. The introduction of Electronic Funds Transfer (EFT), Depository System, Delivery versus Payment (DVP), High-Value Inter-bank Payment System, etc. are essential prerequisites for ensuring a risk-free and transparent payment and settlement system.

vii) Unlike capital markets or commodity markets, trading in the money market is concluded over the telephone followed by written confirmation from the contracting parties. Hence, integrity is *sine qua non*. Thus, banks and other players in the market may have to be licensed and effectively supervised by regulators.

There are not many developed money markets in the world which have all the above-said features. London and New York money markets are the best examples of developed money markers. Other examples of international financial markets are Paris, Zurich, Frankfurt, Amsterdam and Vienna which contain most of the features of developed money markets.

Check Your Progress 1

1) Define the Money Market.

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2) Describe main features of Money Market.

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3) What constitutes a good Money Market?

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10.5 MONEY MARKET IN INDIA

The Indian money market has a dichotomic structure. It has a simultaneous existence of both the organized money market as well as unorganised money markets.

10.5.1 Organised Sector

The *organised* segment is quite integrated and well-organised. The organised money market is in full control of the RBI. In organized markets, there are standardized rules and regulations governing their financial dealings. There is also a high degree of institutionalization and instrumentalization. These markets are subject to strict supervision and control by the RBI or other regulatory bodies. The nature of the money market transactions is such that they are large in amount and high in volume. Thus, the entire market is dominated by a small number of large players. At the same time, the money market in India is yet underdeveloped. The organized money market in India is not a single market but is a conglomeration of markets of various instruments which can broadly be put into the *call money market, the bill market, and other sub-markets*.

Mumbai, Kolkata, Chennai, Delhi, Bangalore and Ahmedabad are the leading centres of the organised sectors of the Indian money market. The Mumbai money market is well organised, having head offices of the RBI and different commercial banks, two leading well-developed stock exchanges, the bullion exchange and a fairly organised market for Government securities. All these have placed the Mumbai money market at par with the New York money market of the USA and the London money market of England.

10.5.2 Instruments of Organised Markets

There are different varieties of money market instruments, but all have a few things in common. In India, the introduction of a host of new instruments [certificates of deposit (CDs), commercial paper (CPs), inter-bank participation certificates, and *money market mutual funds* (MMMF)] have made the Indian money market a vibrant one. The instruments of the money market are characterized by short duration, large volume and deregulated interest rates. The instruments are highly liquid. They also are safe investments owing to the issuer's inherent financial strength.

i) **The call money market**: Call money refers to the unsecured segment of the money market that is designed for the management of liquidity for a very short period — mostly overnight. If the period is more than two days and up to 14 days it is called '*Notice money*'. Money lent for 15 days to 1 year is called 'Term money'. The call market enables banks to even out their day-to-day demand for and supply of short-term funds. This market is a purely unsecured market as no collateral is offered for securing the funds and there are no brokers involved. Settlement is done between the participants through the current account maintained with the RBI. The ease of transacting as well as the low transaction costs arising from the least documentation and same-day settlement of funds act as strong incentives for banks to access the call money market for managing intra-day funding requirements.

The money invested by banks in the call money market provides high liquidity with limited profitability. The call money market in India takes the form of *an inter-bank* call *money market* wherein the banks with surplus lend to the needy banks. Banks usually borrow funds in this market to *meet their statutory obligations* about the keeping of reserves with the Reserve Bank. The rate of interest in this market is highly volatile and sensitive to changes in the demand and supply of funds, with the changes taking place several times during the day. *It is looked upon as the barometer of the money market*. The call money rates also move in certain phases. For example, April, May and October generally witness high rates and high volatility while in January and August, the rates are lower and steady. Various measures have been taken by RBI to reduce volatility in call rates. The setting up of the *Discount and Finance House of India (*DFHI) in 1988 was to broaden the base of the call loan market. DFHI has now assumed the role of *market maker* in the call loan market but due to its inadequate resources, it has not been able to supply enough funds to check volatility.

This is the most important instrument in terms of volume of transactions and in the sense that *rates in this market are indicative of conditions prevailing in the money market*. The rates of other instruments are linked to the rates obtained in the call market. The principal call loan centre is Mumbai which accounts for nearly 85 to 90 per cent of the business transacted. The other centres are *Kolkata,* *Delhi, Chennai, Bengaluru and Ahmedabad*. The major players in this market are commercial banks and DFHI who can both borrow and lend. Select financial institutions like UTI, LIC, IDBI, NABARD etc. have been permitted to operate in the call market as lenders which helps them in effective cash management.

ii) **The bill market**: The bill market can be divided into two: (i) the treasury bills market, and (ii) the commercial bills market.

a) **Treasury bills**: The Union Government raises operational cash by issuing treasury bills. *The Treasury bill* is a short-term government security having a maturity period ranging from a few days to a few months. The instrument is sold by the Central Bank on behalf of the government. Treasury bills are *promissory notes* and at present, these are being issued for 3 maturities viz. (i) 91 days, (ii) 182 days, and (iii) 364 days. Treasury bills are zero coupon securities and pay no interest. Instead, they are issued at a discount and redeemed at the face value at maturity. For example, a 91-day Treasury bill of ₹100/- (face value) may be issued at say ₹ 98.20, that is, at a discount of say, ₹1.80 and would be redeemed at the face value of ₹100/-. The return to the investors is the difference between the maturity value or the face value (that is ₹100) and the issue price. The treasury bills are generally sold by auction to the highest bidder. Being government papers, *the treasury bills enthuse the confidence* of a greater number of investors. Treasury bills are good papers for commercial banks to invest their short-term funds. There is no default risk associated with such instruments. They are *approved assets for Statutory Liquidity Ratio* (SLR). *DFHI is the market maker in these* instruments. This is the most important instrument for hedging against volatility in the call loan market. All T-Bills auctions are Price-based and are auctioned on Multiple-Price basis.

b) **Commercial bill market: bills rediscounting scheme (BRS)**: This scheme has been operative since 1970. Commercial banks are permitted to issue derivative usance promissory notes for maturity not more than 90 days backed by genuine trade bills of similar tenor. This can be further rediscounted by the holder by endorsement and delivery. The stamp duty on this usance promissory note was waived in 1989. *All commercial banks are eligible to issue usance promissory notes* which can be discounted by commercial banks, select financial institutions and urban cooperative banks. DFHI participates actively in this market and provides bid rates daily. This assures liquidity to the instruments. This is a very important source of funds for commercial banks as borrowing under BRS is not included as a liability while calculating Net Demand and Time Liability (NDTL). Due to high liquidity safety and attractive returns, it is a favourable instrument for investors.

iii) Other sub-markets

a) Certificate of deposits (CDs): Banks can raise short-term cash by issuing certificates of deposit. It pays the holder higher interest rates the longer the cash is held. The CD is a document to the title of time deposit but is distinguished from conventional time deposit in respect of its free negotiability. The CD was introduced in the Indian money market in March 1989. The guidelines for their issue have changed over the years in consonance with the development of the money market.

Its term generally ranges from three months to five years and restricts the holders to withdraw funds on demand. However, on payment of certain penalty the money can be withdrawn on demand.

They are freely transferable by endorsement and delivery but only after 45 days of the date of issue. CDs are issued at a discount calculated front end or rear end. The discount rate is market-determined. Initially, only Commercial banks were authorized to issue CDs. In 2021, RBI decided to permit RRBs to issue CDs. It is an *important source for mobilizing bulk funds by banks* at low operational costs. The returns on Certificate of Deposits are higher than T-Bills because it assumes higher level of risk. While buying Certificate of Deposit, return method should be seen. Returns can be based on Annual Percentage Yield (APY) or Annual Percentage Rate (APR). In APY, interest earned is based on compounded interest calculation. However, in APR method, simple interest calculation is done to generate the return. Accordingly, if the interest is paid annually, equal return is generated by both APY and APR methods. However, if interest is paid more than once in a year, it is beneficial to opt APY over APR.

b) **Commercial papers (CPs)**: Large companies with impeccable credit can simply issue short-term unsecured promissory notes to raise cash. CP is a usance promissory note issued by a corporate entity at a discount with a fixed maturity evidencing the short-term obligation of the issuer. CP was introduced in the Indian money market in 1989. The issue of CP seeks to bypass the intermediation process of the banks. It is a low-cost alternative to bank loans. It is a short term unsecured promissory note issued by corporates and financial institutions at a discounted value on face value. They are usually issued with fixed maturity between one to 270 days and for financing of accounts receivables, inventories and meeting short term liabilities. Say, for example, a company has receivables of Rs. 1 crore with credit period of 6 months. It will not be able to liquidate its receivables before 6 months. The company is in need of funds. It can issue commercial papers in form of unsecured promissory notes at discount of 10% on face value of Rs. 1 crore maturing after 6 months. The company has strong credit rating and easily finds buyers. The company is able to liquidate its receivables immediately and the buyer is able to earn interest of Rs. 5 lakhs over a period of 6 months. They yield higher returns as compared to T-Bills as they are less secure in comparison to these bills. Chances of default are almost negligible but are not zero risk instruments. Commercial Paper being an instrument not backed by any collateral, only firms with high quality credit ratings will find buyers easily without offering any substantial discounts. They are issued by corporates to impart flexibility in raising working capital resources at market determined rates. Commercial Papers are actively traded in the secondary market since they are issued in the form of promissory notes and are freely transferable in demat form.

c) **Banker acceptance** is also a money market instrument to meet short-term liquidity requirements. The company provides a bank guarantee to the seller to pay the amount of goods purchased at the agreed future date. In case the buyer fails to pay on the agreed date, the seller can invoke a bank guarantee. It is usually used to finance exports and imports.

d) **Repurchase options (Repo) or ready forward (RF)**: Repo transactions have gained tremendous importance due to their short tenure and flexibility to suit both lender and borrower. Under this transaction, the borrower places with the lender certain acceptable security against funds received and agrees to reverse this transaction on a predetermined future date at the agreed interest cost. *No fixed period has been prescribed for this transaction* however; *generally, repo transactions are for a minimum period of 14 days* and a maximum period of 1 year. *The interest is market-determined and built into the structure of the Repo transaction*. Repo transactions can be undertaken by commercial banks, financial institutions, brokers DFHI. Repo transactions are structured to suit the requirements of both borrower and lender and have, therefore, become an extremely popular mode of raising/investing short-term funds. The borrower has the advantage of raising funds against its securities without altering its asset mix while the lender finds a safe avenue giving an attractive return. Moreover, the fund's management for both borrower and lender is improved as the date of reversal of the transaction is known in advance.

These instruments together have made the Indian money market a vibrant one.

10.5.3 The Unorganized Money Market

The ***unorganized market*** is comprised of indigenous financial institutions like indigenous bankers, lenders, *nidhis* and *chit funds*. They lend money to the public and also collect deposits from the public. There are also private finance companies, chit funds etc., whose activities are not controlled by the RBI. In 1991, the RBI has taken steps to bring private finance companies and chit funds under its strict control by issuing non-banking financial companies (Reserve Bank Directions, 1998). However, the regulations concerning their financial dealings are still inadequate and their financial instruments have not been standardised. The following are some of the constituents of the unorganised money market in India.

i) **Indigenous Bankers**: Indigenous bankers include those individuals and private firms which are engaged in receiving deposits and giving loans thereby acting like a mini bank. Their activities are not at all regulated. With the growth of commercial banks and cooperative banks, the area of operations of indigenous bankers has contracted. These days a few thousand indigenous bankers are operating in the western and southern parts of the country and engaging themselves in the traditional banking business. Indigenous bankers are classified into four main subgroups, i.e., *Gujarati Shroffs, Multani-or Shikarpuri Shroffs, Chettiars and Marwari, and Kayast. Gujarati Shroffs* are mostly operating in Mumbai, Kolkata and industrial and trading cities of Gujarat. The *Multani or Shikarpuri Shroffs* are operating mainly in Mumbai and Chennai. The *Chettiars* are mostly found in the South. The *Marwari Shroffs* are mostly active in Mumbai, Kolkata, the tea gardens of Assam and also in different other parts of North-East India. Among the four aforementioned groups, the Gujarati indigenous bankers are considered the most powerful groups in respect of their volume of business.

   The indigenous bankers are mostly engaged in both banking and non-banking business which they do not want to separate. Their lending operations remain mostly unregulated and unsupervised. They charge a high rate of interest and they are not influenced by the bank rate policy of the RBI.

ii) **Unregulated Non-Bank Financial Intermediaries**: There are different types of unregulated non-bank financial intermediaries in India. They are mostly constituted by loan or finance companies, *chit funds* and '*nidhis*'. A good number of finance companies in India are engaged in collecting substantial amounts of funds in the form of deposits, borrowings and other receipts. They normally give loans to wholesale traders, retailers, artisans, and different self-employed persons at a high rate of interest ranging between 36 to 48 per cent.

   There are various types of chit funds in India. They are doing business in almost all the states but the major portion of their business is concentrated in Tamil Nadu and Kerala. Moreover, there are '*nidhis*' operating in South India which are a kind of mutual benefit fund restricted to its members.

iii) **Moneylenders**: Moneylenders are advancing loans to small borrowers like marginal and small farmers, agricultural labourers, artisans, factory and mine workers, low-paid staff, small traders etc. at very high rates of interest and also adopt various malpractices for manipulating loan records of these poor borrowers. There are broadly three types of moneylenders:

   i) Professional moneylenders dealing solely with money lending;

   ii) Itinerant moneylenders such as *Kabulis and Pathans*, and

   iii) Non-professional moneylenders.

The area of operation of the moneylenders is very much localised and their methods of operation are also not uniform. The money lending operation of the moneylenders is unregulated and unsupervised which leads to the worst exploitation of the small borrowers. Moneylenders have become a necessary evil in the absence of sufficient institutional sources of credit for the poorer sections of society. Although various measures have been introduced to control the activities of moneylenders due to a lack of political will, these are not enforced, leading to the exploitation of small borrowers.

Check Your Progress 2

1) List the main constituents of the Indian Money Market.

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2) Distinguish between Certificate of deposits (CDs) and commercial papers (CPs).

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3) Distinguish between Indigenous Bankers and Moneylenders.

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4) Explain the importance of treasury bills.

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10.6 REGULATORY MEASURES TO STREAMLINE THE WORKING OF THE MONEY MARKET

In India, money markets are regulated by both RBI and SEBI. The Indian money market has undergone metamorphosis during the last two decades owing to a series of measures which increased the number of participants, introduced newer instruments and deregulated the interest rates. RBI undertook several policy measures for setting up *institutions* to accelerate the growth and development of the Indian money market. The growth of the Indian money market is gaining momentum and is getting interrelated with other markets.

Three important committees were set up. While the *Chakravarty Committee* (1985) recommended measures for improvement in the monetary system, the *Vaghul Working Group* (1986) recommended measures to activate and vitalize the money market and the *Narasimham Committee* (1991) recommended measures to streamline the functioning of the financial system. Notwithstanding the above, it was only after the initiation of the reform measures that the money market gained vibrancy, although not to a very high degree. However, a significant aspect of the reform measures has been that the financial system as a whole, has become more inter-related and today the impact of the volatility in one market is felt in the other market. Keeping in mind the need for a developed money market for a fast-growing economy like India, RBI undertook several policy measures for setting up *institutions* to help hasten the growth and development of the Indian money market. Accordingly:

i) *The Discount and Finance House of India (DFHI)* was established in 1988 to broaden and deepen the money market operations. It acts as a market maker;

ii) *The Securities and Exchange Board of India (SEBI)* was set up in 1992 to function as the apex regulatory body relating to all aspects of security market activities;

iii) *Credit Rating agencies such as CRISIL* (1987)*, ICRA* (1991), etc. were promoted to make possible easy raising of funds from the money market.

Considering the need for making available an adequate number of and different variety of financial instruments as a precursor for a developed money market, several instruments were introduced in the Indian money market. *Several innovative submarkets such as certificates of deposit, call money market, collateral loan market, commercial paper market; MFS, etc. come to be introduced to strengthen the money market*. A striking feature of the money market instruments is that the *interest rate structure applicable to them has been deregulated* with effect from May 1, 1989. Accordingly, the rates of interest will be market-determined. The various money market instruments that are mentioned above have some of the following characteristics: *short maturity, high liquidity, high safety, and high volume.* An essential feature of these instruments is that they are being dealt with by established agencies such as RBI, DFHI and Securities Trading Corporation of India (STCI) at the apex level and by commercial banks, public financial institutions and selected corporate entities at the operating level. Whereas commercial banks can indulge in both buying and selling of these instruments without any restriction, only lending facilities with the fulfilment of certain conditions are available for selected public financial institutions and corporate entities.

Credit control measures such as CRR and SLR are efficiently manipulated to tone up and buoy the money market operations, especially in the commercial banking sector. The reintroduction of *repos (ready forward*) in November 1996, was aimed at mopping up excess liquidity thus reducing volatility in call money markets and on foreign exchange markets. Beginning October 21, 1997, the repo facility was further extended to corporate debt and PSU bonds. Under this arrangement, RBI made *reverse repo facility* available to primary dealers in the government securities market, at the bank rate, on a discretionary basis and subject to certain regulations from time to time.

The Indian money market also has attained greater *vibrancy over* the years. Since the initiation of the liberalisation measures in mid-1991, there has emerged a *stronger relationship between the money markets and the other markets* viz., the foreign exchange market, capital market and the central government securities market, in the financial system. For all this to happen, the necessary policy changes carried out through the credit policies of the RBI have played a very significant role.

Money market rates play a main role in controlling the inflation. Higher rates in the money markets decrease the liquidity in the economy and have the effect of reducing the economic activity in the system. Reduced rates on the other hand increase the liquidity in the market and bring down the cost of capital considerably, thereby raising the investment. This function also assists the RBI in controlling the general money supply in the economy. To further strengthen the regulatory architecture while promoting greater efficiency in the financial markets, the Reserve Bank initiated policy measures to address certain regulatory gaps, promote transparency and improve liquidity in the money markets.

10.7 LIMITATIONS OF INDIAN MONEY MARKET

In terms of both organisation and development, the Indian money market is not comparable to any of the developed money markets of the world. It also cannot match the extent of resources, stability and elasticity of these developed money markets. But among the third-world countries, India has been maintaining the most developed banking system. Even then the organisation of the money market suffers from several drawbacks.

There is no adequate and continuous supply of short-term assets such as treasury bills, bills of exchange, short-term Government bonds etc. Accordingly, there are no dealers of short-term assets who act as intermediaries between the Government and the entire banking system. Despite the serious efforts made by the Reserve Bank of India, the bill market in India has not yet been fully developed. The market suffers from a lack of coordination between its different constituents and fails to attract foreign funds. Because of the existing dichotomy between organised and unorganised sectors, it is difficult for the RBI to ensure uniform and effective implementations of monetary policy in both sectors. Wasteful competition exists not only between the organised and unorganised sectors but also among the members of the two sectors. As compared to the size and population of the country, the banking institutions are not enough. Inadequate banking facilities in vast rural areas still exist. Disparities in the interest rates adversely affect the smooth and effective functioning of the money market. In recent years, serious efforts have been made by the Government and the RBI to remove the shortcomings of the Indian money market. Moreover, with the setting up of DFHI and MMMFs, the lot of Indian money market has achieved considerable progress in recent times and is also expected to achieve further progress in the years to come.

Check Your Progress 3

1) For what purposes three committees were set up?

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2) What reforms were introduced by RBI to strengthen the money market in India?

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3) Highlight the limitations of the Indian Money Market.

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10.8 LET US SUM UP

The money market is an organized exchange market where participants can lend and borrow short-term, high-quality debt securities with average maturities of one year or less. The market enables governments, banks, and other large institutions to sell short-term securities to fund their short-term cash flow needs. It also allows individual investors to invest small amounts of money in a low-risk market. The short period of maturity is the hallmark of the money market instruments, while the markets that cater to the need for long-term funds are called Capital Markets. The nature of the money market depends to a large extent on the kind of economy it is expected to serve.

The money market instruments are money at call and short notice (Call Loans). Treasury Bills (TBs), Bills Rediscounting Scheme (BRS), Certificates of Deposit (CDs), Commercial papers (CPs), Repurchase Options or Ready Forward (REPOs), Options, Financial Futures, Forward Rate Agreements, Swaps, etc.

The instruments of the money market are characterized by short duration, large volume and deregulated interest rates. The instruments are highly liquid. They also are safe investments owing to the issuer's inherent financial strength.

New instruments have evolved and more institutions have been allowed to participate and market makers have been created to impart depth to the market. The money market in India consists of formal or organized money markets and informal or unorganized money markets.

In a money market, the central bank of the country plays a pivotal role and occupies a strategic position. The central bank is the key constituent of the money market, being the residual source of supply of funds and this attaches great significance to its operations.

The objective of monetary management by the central bank is to align money market rates with the key policy rate. As excessive money market volatility could deliver confusing signals about the stance of monetary policy, it is critical to ensure orderly market behaviour, from the point of view of both monetary and financial stability. Thus, the efficient functioning of the money market is important for the effectiveness of monetary policy.

10.9 KEY WORDS

Bill of Exchange : A negotiable instrument with three parties to the agreement: the drawer of the order, the drawee, who is ordered to make payment, and the payee. For instance, a commercial bank cheque is a draft on the bank by the depositor payable to a designated payee.

Credit Controls : Regulatory restrictions on bank lending.

Financial Integration : How financial markets are tied together geographically — domestically and internationally.

Market Maker : Refers to the dealers who buy or sell securities from their inventories, thereby ensuring that there is always a market in which investors can buy or sell their securities.

Monetary Policy : The management of the money supply and its links to prices, interest rates and other economic variables.

Open Market Operations : The purchase and sale of securities in financial markets by the central bank.

Regional Rural Banks (RRBs) : RRBs were established in 1975 under the provisions of the Regional Rural Banks Act. 1976 to develop the rural economy.

Statutory Liquidity Ratio (SLR) : Refers to that portion of total deposits of a commercial bank which it has to keep in the form of government or government approved securities.

Underwriters : Underwriters are those intermediaries who take upon themselves the responsibility to subscribe to that portion of the securities issued by a company which remain un-subscribed by the investing public.

10.10 SOME USEFUL BOOKS

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- Gupta, Suraj B. (2010): Monetary Economics; Institutions, Theory and Policy, S. Chand and Company, New Delhi (Chapters – 4, 18 & 19).
- Khan, M Y (2017): '*Indian Financial System*', 10th edition, Chapter 9, Tata McGraw-Hill Education Pvt. Ltd., New Delhi
- Mohanty, Deepak (2013): Money Market and Monetary Operations in India, *RBI Bulletin*, January
- Report of the Committee on Banking Sector Reforms (Chairman: M . Narsimham)1998
- Report of the Working Group on Money Market (Chairman: N. Vaghul), 1987
- Reserve Bank of India (2011): Reserve Bank of India: Functions and Working, RBI, Mumbai (Various issues)
- Reserve Bank of India: Report on Currency and Finance. Annual Report (Various issues)

10.11 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Section of the financial market where financial instruments with high liquidity and short-term maturities that usually range from overnight to just under a year such as treasury bills and commercial papers. It is considered a safe place to invest due to the high liquidity of securities.

2) Consists of negotiable instruments, provides an avenue for equilibrating the short-term surplus funds of lenders and the requirements of borrowers, collection of markets, such as, call money, notice money, repose, term money, treasury bills, commercial bills, certificate of deposits, commercial papers, inter-bank participation certificates, inter-corporate deposits, swaps futures, options, etc. volume of funds or financial assets traded in the money market is very large.

3) Highly organised commercial banking and financial systems, Promote financial mobility, Continuous supply of negotiable securities, Existence of several sub-markets, integrated with the rest of the markets in the financial system.

Check Your Progress 2

1) Organized money market, unorganised money markets.

2) The CD is a document to the title of time deposit but is distinguished from conventional time deposit in respect of its free negotiability. Banks can raise short-term cash by issuing certificates of deposit. CP is a usance promissory note issued by a corporate entity at a discount with a fixed maturity evidencing the short-term obligation of the issuer. *The issue of CP seeks to bypass the intermediation process of the banks.*

3) Indigenous bankers include those individuals and private firms which are engaged in receiving deposits and giving loans thereby acting like a mini bank. Moneylenders are advancing loans to small borrowers like marginal and small farmers, agricultural labourers, artisans, factory and mine workers, low-paid staff, small traders etc. at very high rates of interest.

4) *The Treasury bill* is a short-term government security having a maturity period ranging from a few days to a few months. There is no default risk associated with such instruments. They are *approved assets for Statutory Liquidity Ratio* (SLR).

Check Your Progress 3

1) Three important committees were set up. While the *Chakravarty Committee* (1985) recommended measures for improvement in the monetary system, the *Vaghul Working Group* (1986) recommended

measures to activate and vitalize the money market and the *Narasimham Committee* (1991) recommended measures to streamline the functioning of the financial system.

2) The establishment of *the Discount and Finance House of India (DFHI)(1988), the Securities and Exchange Board of India (SEBI)(1992), Credit Rating agencies such as CRISIL* (1987)*, ICRA* (1991).

3) Inadequate and discontinuous supply of short-term assets such as treasury bills, bills of exchange, short-term Government bonds etc., no dealers of short-term assets who act as intermediaries between the Government and the entire banking system, lack of coordination between its different constituents, failure to attract foreign funds.

10.12 TERMINAL QUESTIONS

1) Explain the constituents and salient features of the Indian money market.

2) Discuss various characteristics of a good money market. Can the Indian money market be termed a developed money market? Outline the measures to improve the functioning of the Indian money market.

3) What reform measures RBI has adopted to diversify and strengthen the various instruments of the money market?

4) Examine the role and importance of RBI in the functioning of the Indian money market.

UNIT 11 CAPITAL MARKETS

Structure

11.0 OBJECTIVES

After going through this unit, you should be able to:

- Describe the purposes for which capital markets exist and the role they play in a modern economy;
- List the various participants and the types of exchanges taking place in these markets;
- Compare between debt and equity as means of raising finance for business firms;
- Analyse the working of debt markets, the instruments in these markets and the pricing in these markets;
- Discuss the principles of the working of equity markets; and
- Explain the concept of share price indices and the methods of their construction.

11.1 INTRODUCTION

The previous unit acquainted you with markets for short-term funds, and we learnt that these markets are called money markets. The present unit deals in detail with various markets for long-term funds. These are called capital markets. We will see there are primarily two types of capital markets: debt markets, and equity markets. In the next section, we consider the purpose of capital markets in general, and the various participants in these markets. In the section after that, we will look at debt and equity as alternative means of raising finance. Section 11.4 limits itself to the debt market but provides a detailed discussion about the workings of these markets. The subsequent section turns to equity markets and explains why there is volatility, or risk, in these markets. Finally, section 11.6 discusses the indices of share prices and how these are constructed. Let us begin by stating the purpose and pinpointing the uses of capital markets.

11.2 PURPOSE AND USES OF CAPITAL MARKETS

Investment is nothing but postponed consumption. So, there are investors who are willing to lock in their funds in the expectation of earning higher returns and making capital gains. On the other side, there are business firms, organisations, and individuals who need funds and finance for making physical investments in the hope of making profits or gains, or even for consumption. Capital markets constitute a mechanism for bringing together buyers and sellers of various types of financial assets. Exchanges of these financial assets determine their prices.

As we have discussed in previous units the financial system of a nation consists of financial institutions, financial markets, and financial instruments and services. A financial market is an arrangement that facilitates the exchange of financial assets, including deposits and loans, corporate stocks and bonds, government bonds, and derivatives like futures and options. You will read about derivatives in the next unit. Financial markets can be classified in several ways: as primary and secondary markets, that is markets in new issues and markets in existing issues; as organised or formal markets, and as unorganised or informal markets. Financial markets can also be classified as money markets, markets for short-term funds, and capital markets, or markets for long-run funds.

Capital markets can be markets where long-term debt contracts are traded, like bond markets or debentures, or these can be equity markets where financial claims are sold which allows the holder a share in the profits of the firms issuing these shares. The markets where shares are issued initially are called primary markets, and the markets where the holders of these shares further trade these in the expectation of making capital gains are called secondary markets.

Let us look now at the basis for the need and provision of finance, as this will enable us to know about debt as the foundation of finance, and see who the participants in the markets are. Capital markets are all about the raising of capital and matching those who want capital (borrowers) with those who have capital (lenders). One way of matching these are financial institutions like banks and other depository institutions. For more complex transactions, markets are needed in which borrowers (and their agents) can meet lenders (and their agents). Commitments to borrow and lend are created. There are also markets where these commitments to borrow and lend are themselves sold off to other people. For example, companies can raise money by selling shares to investors and existing shares can be freely bought and sold.

Who are the borrowers? Individuals can borrow for domestic purchases or longer term such as financing and mortgages of houses, consumer durables, or education... companies need short-term funds to finance their cash flow. They need longer-term funds for growth and expansion. Governments are huge borrowers. They often borrow by issuing securities like bonds. Their total borrowing is called National Debt. Public sector companies may also borrow for expansion and growth. Borrowing may also be done from foreign agencies, institutions etc. Similarly, lenders may be individuals, financial institutions, or even companies. A company can use some of its idle funds to lend for short periods to make gains, in the money market.

When money is lent, in many cases the borrower issues a receipt for the money, a promise to pay back. These receipts are called securities, and the lender has claims on these securities. Thus, securities are financial claims. The holders of financial securities are the lenders, and the issuers of financial securities are the borrowers. There are many types of securities like Treasury Bills, Certificates of Deposits, Commercial Paper, bonds, debentures, equity shares, and so on. All securities show key information about how much is owed, when it will be paid back, and how much is the rate of interest.

So, we have seen that markets exist for finance and the main participants are borrowers, lenders, their agents and other individuals and agencies that facilitate exchanges. Borrowers issue securities. Lenders purchase these securities. For example, when the government issues a bond, we say that the government is selling the bonds. When you purchase the bond, you are lending money to the government, and the government is borrowing money from you. The market for long-term securities is called a capital market.

Capital markets perform several important functions in a market economy. The capital markets make information processing and dissemination easier, which allows better decisions can be taken regarding investment, selling of assets, and so on. Capital markets allow quick valuation of financial instruments – both debt and equity. Capital markets also enable operational efficiency by simplifying transaction procedures and lowering search costs and transaction costs. Capital markets help to develop integration among several sectors of the economy such as between real and financial sectors, between debt and equity instruments, between short-term and long-term funds, and between domestic and external funds. Finally, capital markets facilitate the flow of funds into efficient channels through proper investment.

11.3 DEBT AND EQUITY AS MEANS OF RAISING FINANCE

In this section, we will take a look at one particular borrower: the business firm. We shall see how firms make decisions about raising finance for their short-term and long-term requirements. Firms can raise finance either by borrowing (debt) or by issuing shares. So when we are talking about shares, we have only the primary market in mind. We are looking from a firm's point of view and asking how much of the finance should be raised by incurring debt, and how much should be raised through the issue of shares. Is there some theory or some principles that will help the firm to make a decision? This is what we are discussing now.

A firm has several assets that it invests in, and these investments generate a stream of cash flows. If the firm is entirely equity financed, these cash flows belong only to the shareholders. On the other hand, the firm can offer shares but also go in for borrowings. This will offer the lenders some risk-free investment opportunities. How much of the capital the firm should raise through debt and how much through equity is a decision that is important for the firm. This is called the **capital structure** of the firm. To understand the decision regarding the capital structure of the firm, we will first look at some indicators of the firm's financial position, particularly some financial ratios. One can look at the balance sheet and the income statements. Balance sheets look at the situation with assets and liabilities at a particular point of time while the income statement is a view of the earnings and expenditure that takes place over a specified period, usually a year. We can also look at a firm's generation of cash. But what we propose to do now is to look at some important financial ratios regarding a firm's financial activity and performance. Ratio analysis forms a powerful tool for assessing the financial condition of the firm and enables us to relate disparate financial data with each other. Ratio analysis of this sort will enable us to understand the capital structure of the business firm.

Different people and groups have varying stakes in the firm, and they will have different concerns over a firm's performance. Stockholders and bondholders will be concerned about the long-term profitability of the firm, while creditors will look to see whether the firm can meet its payments on time. The firm's management is concerned with both the long run and short run as well as day-to-day performance. There are several types of ratios that we shall be considering now. All of these can be grouped into four types: **liquidity ratios**, which measure the ability and adequacy of the current assets to meet current liabilities as they come due; **activity ratios**, which indicate the efficiency with which the firm uses its resources; **financial leverage ratios**, which measure how effective the firm is in managing its debt; and finally, **profitability ratios**, which measure how effective the firm is in generating net revenues and income about sales, costs, assets and shareholder equity.

Let us begin with liquidity ratios. **Liquidity ratios** refer to a firm's ability to meet its short-term obligations and are concerned with the size and composition of the firm's working capital position. A higher working capital position implies more liquidity. An important ratio is the **current ratio,** which equals current assets divided by current liabilities. This ratio indicates the amount of current assets available to meet all its obligations under current liabilities. The current ratio includes inventories, but since inventories are among the least liquid of a company's assets, it is sometimes useful to exclude them. Deducting inventories from the current ratio gives us the **quick** or **acid-test ratio**:

$$\text{Quick ratio} = \frac{(\text{Current assets} - \text{Inventories})}{\text{Current Liabilities}}$$

Next, we turn to some activity ratios. These by and large have to do with sales generated, often about the amount invested in them. One such measure is the **accounts receivable turnover**, which equals net credit sales divided by turnover. A related measure is the average collection period, which is given by (365) (receivables)/ (net credit sales). We can look at the total asset turnover which is an overarching activity ratio and is defined as:

$$\text{Total asset turnover} = \frac{\text{Net sales}}{\text{total assets}}$$

Now let us focus on some financial leverage ratios, which will enable us to talk about the capital structure of firms. Sometimes, as in Britain, leverage is called 'gearing'. Leverage ratios reflect a firm's risk position by analysing its financing mix. The idea is that the more the operations of the firm are financed through debt, the greater the risk. Information on this can be obtained from the firm's balance sheet. The other way is to use income statement data to derive the coverage ratios, which measure the company's ability to service that debt. Two widely used ratios derived from the information in balance sheets are the debt ratio and the debt-equity ratio. By debt, here we mean the firm's current liabilities as well as long-run debt. The **debt ratio** is defined as:

Debt ratio = Total Liabilities divided by Total Assets

The **debt-equity ratio** is defined as:

Debt-equity ratio = Total Liabilities divided by Equities

Now since total assets equal total liabilities plus equity, it can be shown (it is left to you as an exercise) that:

Debt-equity ratio = Debt ratio divided by (1 – Debt Ratio)

From the income statements we can derive the firm's cash flow coverage, which is given as

$$\textbf{Cash Flow Coverage} = \frac{\text{EBIT} + \text{Lease/rental Payments} + \text{Depreciation}}{Interest + Lease/Rental\ Payment + Principal\ Repayment/(1-T)}$$

Where EBIT stands for annual earnings before interest and taxes, and T stands for taxes on a firm.

Finally, we discuss profitability ratios. These ratios allow the shareholders to evaluate management's performance. There are two types of profitability ratios used: profit margin on sales and returns on assets employed. Profit margins — a type of profitability ratio that looks at a firm's expenditure on sales. There are two important such ratios: gross profit margin and net profit margin. They are defined as follows:

$$\textbf{Gross Profit Margin} = \frac{\text{Gross Profits}}{\text{Net Sales}}$$

$$\textbf{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

In addition, we can work out the **operating profit margin which is defined as:**

$$\textbf{Operating profit margin} = \frac{\text{EBIT}}{\text{Net Sales}}$$

In addition to looking at ratios related to the profit margin, we can relate returns to assets and shareholders' equity. Two ratios frequently used are **(ROA)**, also known as **return on investment (ROI)**, and **return on equity (ROE)**. They are defined as:

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Total Assets}}$$

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

Equipped with these ratios, in the next section, we shall discuss, somewhat briefly, some elements of the capital structure of a firm.

Check Your Progress 1

1) What are the main functions performed by capital markets in the economy?

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2) Name and define two liquidity ratios and two financial leverage ratios.

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11.4 DEBT MARKET INSTRUMENTS AND THEIR PRICING

The main debt instruments are bonds, which are fixed-income securities. The investor (who is the lender) gets periodic interest payments over the life of the bond and the principal payment at the time of redemption of the bond. The issuer of the bond (who is the lender, that is who undertakes to bear the debt) promises to pay the holder (buyer) of the bond this periodic stream of payments and the principal at the end. Debentures are like bonds, except that there is a clause that enables the investor to convert the debenture into equity shares on certain pre-specified terms and conditions.

There are short-term debt instruments, which you studied in detail in the previous unit on money markets, like treasury bills, certificates of deposits, commercial paper etc. In this unit, we limit ourselves to a study of bonds. Bonds are issued by the government and these are government-guaranteed. For example, in India, the Government of India is the largest borrower. The government sells short-term to medium-term bonds issued by the Reserve Bank of India. State governments also sell bonds. Apart from the central and state governments, several government agencies issue bonds that are guaranteed by the government. Other than government bonds, there are bonds issued by public-sector companies, financial institutions, and private-sector companies.

Over the last couple of decades, there have been several innovations in the types of bonds that have been issued. Traditionally simple bonds have been issued (these are known as 'plain vanilla bonds' or straight bonds). These bonds usually pay a fixed periodic coupon over its life and return the principal on the maturity date. The new types of bonds are of several types: zero-coupon bonds; floating rate bonds; bonds with embedded options; and commodity-linked bonds. These are explained below:

Zero coupon bonds do not give regular interest payments. It is issued at a large discount and redeemed at face value on maturity. Sometimes, these bonds carry the call and put options. You will know about put and call options in the next unit. Floating rate bonds, unlike straight bonds that pay a fixed rate of interest, pay an interest rate that is linked to a benchmark rate such as the interest rate on Treasury bills. Bonds with embedded options include bonds like debentures that give the bondholder the right to convert them into equity shares; as well as callable bonds, which are bonds that give the issuer the right to redeem them prematurely on certain terms; and bonds with put options, which are bonds that give the investor the right to prematurely sell them back to the issuer on certain terms. Finally, there are commodity-linked bonds that are bonds the return from which depends on the price of certain specified commodities.

We will be looking at the risks associated with bonds in detail in unit 13 of Block 4. For the present, we proceed from the assumption that bonds are risk-free. With that assumption, let us now discuss the value of a bond and see how the value of a bond is determined. The value of a bond is equal to the present value of the expected cash flows arising from it. Here you will do well to go over unit 4 in Block 1. Hence we need to look at the expected stream of cash flows and an estimate of the returns. We assume further that the bond pays a fixed coupon interest rate, that coupon payments are made annually and that the bonds will be redeemed at par when it matures.

With these assumptions, the value for such a bond is:

$$A = \sum_{t=1}^{T} \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

where A = value (in rupees)

T = number of years

X = annual coupon payment (in rupees)

The required return is denoted by r

M = maturity value

The period when the payment is received is denoted by t.

A basic property of a bond is that there is an inverse relationship between the price of the bond and yield. This is because as the yield increases, the present value of the cash flow decreases; therefore the price falls. What is the relationship between the bond price and time? Since the bond price must equal its par value at maturity (we assume there is no risk of default) bond prices change with time. If the current price prevails when the redemption is higher than the par value of the bond, the bond is said to be a premium bond; if the current price at the time of redemption is lower, the bond is said to be a discount bond.

The trading of bonds is usually based on their prices. But since there are significant differences in cash flow patterns, and also other features, of different bonds, different bonds are not usually compared in terms of their prices. Instead, the point of comparison is bond yields. We now turn to some measures of bond yields. A common measure of yield is the **current yield**, which relates the annual coupon interest to the market price. It is expressed as:

Current Yield = Annual Interest / Price.

The current yield does not consider the capital gain (or loss) that the investor will face if she holds the bond till maturity and sells at a premium (or discount). This measure also neglects the time value of money. The next measure of yield that we consider is the **yield to maturity**. This is the value of the rate of interest that makes the present value of the cash stream received from owning the bond equal to the price of the bond. It can be expressed as

$$P = \sum_{t=1}^{T} \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

Where P is the price of the bond, and the other letters denote the same variable as in the previous equation. Sometimes the calculation of r in the above is difficult and a trial-and-error procedure is resorted to. The final measure of yield that we consider is the **yield to call**. It is calculated in the same way as yield to maturity, except that in yield to call, instead of maturity value at the end, there is call price and the duration, instead of going from say, 1 to T, goes from 1 to the number of years until the assumed call date.

Check Your Progress 2

1) What is yield to maturity?

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2) How do we calculate the expected stream of cash flows arising from bonds?

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11.5 EQUITY: MARKETS AND VOLATILITY

Before we go on to discuss equity markets, let us discuss the role that equity markets play in the economy. We shall use the terms stock market and stock exchange interchangeably. What does a stock exchange do? It provides the regulation of company listings, a price formation mechanism, the supervision of trading, authorisation of members, settlement of transactions and publication of trade data and prices.

Equity markets consist of primary markets and secondary markets. The primary market is the market for new issues of shares, where shares are initially offered to the public. The secondary market, on the other hand, is the market for already-issued financial assets. The main difference between the two markets is that in the secondary market, the issuer of the share does not receive funds from the buyer. Rather, the existing issue changes hands in the secondary market, and funds flow from the new buyer of the share to the original buyer.

Equity capital is ownership capital. It is the shareholders who collectively own the company. They undertake the risks but also earn the rewards of ownership. The amount of capital that a company *can* issue is the **authorised capital**. The amount of capital that a company offers to investors is the **issued capital**. The part of capital subscribed by investors is called the **paid-up capital** assuming that no calls are outstanding i.e., all the shareholders have paid their called-up money. The par value is the face value of the share. **Face value** is the equity share capital divided by the number of equity shares. When the issue price is greater than the par value, the difference is called the **equity premium**. The **book value** (sometimes called the intrinsic value) of an equity share is equal to:

(*paid-up capital + reserves and surplus*) /(*number of equity shares outstanding*). Alternatively, the book value can be found out by dividing the *net assets (assets minus liabilities)* by the number of equity shares outstanding. The **market value** of an equity share is the price at which it is traded in the market.

We now turn to some valuation ratios. Valuation ratios indicate how investors assess the stock in the equity markets. These are thus indicators of the performance of the issuing firm. The important valuation ratios are the price-earnings ratio, the yield, and the ratio of market value to book value.

Price-earnings ratio: It is defined as the market price per share divided by earnings per share. The earnings per share ratio is just the after-tax profit less preference dividend divided by the number of outstanding equity shares. In practice, this ratio is popular with the terminology PE ratio. Higher PE ratio shows higher earning to an equity share which may increase demand of the share.

The price-earnings ratio reflects risk characteristics, investor sentiments, the degree of liquidity, and growth prospects.

Yield: This measures the rate of return earned by shareholders. It is defined as:

$$\text{Yield } = \frac{\text{Dividend } + \text{ price change}}{\text{Initial price}}$$

We can break this into two parts:

$$\frac{\text{Dividend}}{\text{Initial price}} + \frac{\text{price change}}{\text{Initial price}}$$

The first term denotes the dividend yield, while the second term is the capital gains or losses yield. Usually, companies with low growth prospects offer a high dividend yield while the capital gains yield is low. It is the converse about companies with high growth prospects.

Market Value to Book Value ratio: This is defined as the ratio of market value per share to the book value per share. It is a measure of the firm's generation of wealth. If it is more than 1, that is, if the market value exceeds the book value, it means wealth has been generated. We can see that if the ratio of market value to book value is equal to one, all three ratios, namely, earnings price ratio (reciprocal of price-earning ratio), yield and return to equity will be equal to each other.

Nobel laureate James Tobin has proposed a valuation ratio, called Tobin's q, after the letter he used to denote it. The q ratio is defined as:

$$\frac{\text{Market value of equities and liabilities}}{\text{estimated replacement cost of assets}}$$

The q ratio looks a little like the market value to book value ratio. But there are two important differences: the numerator of the q ratio includes liabilities of the issuer of debt instruments along with the market value of equities, that is, it includes all wealth. Also, in the denominator of the q ratio, the assets are valued at their replacement cost, not book value.

The main functions of the secondary markets are that it provides useful information both to the issuer as well as to investors, the function it serves for the issuer is as follows. The secondary market provides regular information, by the latest prices of shares, to the original issuer of the shares about the value of its shares. The periodic trading of the asset reveals to the issuer the price the share commands in the market at a certain time. Such information helps the issuer to evaluate how well they are using the funds from earlier primary market activities, and it also provides some idea about how receptive investors will be to new issues.

To the original buyer of shares in the initial public offering in the primary market, the secondary market acts like an escape clause. If the shares do not seem to be doing as well as the original buyer thought, he or she can always sell them for cash. Of course, the investor can also sell it for cash to make capital gains. The secondary market provides the original buyer of the shares an opportunity to sell these for cash. If the investors are not confident that they can move from one financial asset to another, they might be reluctant to invest in any security. This will make it difficult for the issuers to issue new shares, and they will have to offer higher returns or dividends to attract investors.

The secondary markets also perform a useful function. They provide information about the assets' fair price, as well as liquidity from their assets should they want to sell the assets. Another very important function performed by secondary stock markets is that they reduce search costs, by making it easier for buyers and sellers of assets to find each other. The secondary markets also lower transaction costs.

Let us now discuss some basic parameters that enable one to gauge the functioning of the stock market as a whole, which makes it easier to know if the stock market is developed and mature or not. The parameters that we should look at to judge the functioning of equity markets are volatility, liquidity, size of the market, and transaction cost. **The volatility** of a stock is a measure of the frequency with which its price changes over some time. If the price changes frequently, it means that the stock is volatile. Investors tend to avoid shares with high volatility because of the risks involved. The volatility of stocks also influences the outside flow of funds and hence has a macroeconomic dimension. Proximate factors like speculation and trading and settlement systems, as well as other factors like government policies, interest rates, inflation, and the regulatory framework for equity markets all affect volatility in the stock market.

Liquidity is a very important indicator of the proper functioning of stock markets. The situation about liquidity greatly influences stock market efficiency and development. A market is considered liquid when large volumes of trade can take place without a significant effect on stock prices. If an investor can transact at a price close to the current market price, the market is considered very liquid. There are two methods for measuring liquidity: the turnover ratio and the value traded ratio. The **turnover ratio** equals the ratio of the total value of domestic shares traded to total market capitalisation. This ratio seeks to measure the trading of domestic equities in domestic markets relative to the size of the market. The more the volume of sales, the higher the turnover. High turnover is often an indicator of low transaction costs. Even if a market is large, turnover may be low despite huge market capitalisation if the volume of trading is low, which can occur if the market is not very active. The other measure of liquidity is the **value-traded ratio**. This equals the ratio of the total value of domestic shares traded on the major stock exchanges to the GDP. It is a reflection of organised trading of shares as a proportion of GDP and is thus a macro-level indicator of liquidity in the capital markets.

The third parameter regarding the functioning of equity markets is the size of the stock market, and a useful measure of this size is the **market capitalisation ratio**, which is the ratio of market capitalisation (the value of listed shares i.e., market price × number of the equity shares) to the GDP. It is economically very significant as it is based on the assumption that the size of the stock market is positively correlated with the ability to mobilise capital and diversify risk. Hence it is expected that in an economy with well-developed stock markets, the market capitalisation ratio will be large. This ratio complements the value-traded ratio. The final indicator of the development, maturity and vibrancy of stock markets is the level of transaction costs. In the context of stock markets transaction costs would include explicit costs like brokerage costs, stamp duty, commission etc., as well as implicit costs like clearing and settlement costs, bad delivery, paperwork and so on. If transaction costs are low, investors will be induced to trade in greater volume, which in turn will raise the value traded ratio.

11.6 INDICES OF SHARE PRICES

The stock market index is a very important indicator of the performance of share prices. It takes a set of stocks as representative of the prices of shares in the market, and by tracking the changes in this index, we can get an idea of the overall market sentiment. The stock market index is a pointer to market behaviour. The stock market index reflects market direction and indicates day-to-day fluctuations in stock prices.

What are the characteristics of a good index of share prices? A good index should have two characteristics. First, it should consider a set or group of scrips that have high market capitalisation, secondly, these scrips should also be highly liquid. Market capitalisation, as we have seen, refers to the sum of the market value of all the stocks included in the index. The market value is obtained by multiplying the price of the shares by the number of shares outstanding. Liquidity refers to the ease with which a share can be bought or sold at a price close to the current market price.

The share price index is calculated for a particular day as a percentage of the aggregate market value on that day of the set of stocks included in the index to the average market value of the same shares during the base period, for example, the BSE Sensex (sensitive Index) is a weighted average of the prices of 30 select stocks. Suppose the index is the set of prices of the largest 25 companies, it means that these companies are largest in terms of market capitalisation. Sometimes the index is weighted. If the index is based on the price of, say 35 companies, a weighted index implies that a 1% change in the price of the largest companies in the group constituting the index will have more impact on the index than a 1% change in the price of the smallest. Since share prices are changing frequently, the companies with the top shares do not remain the same and hence the constituents of the index can change.

We must be careful to make a distinction between price averages and price indices. In 1884, for example, Charles Dow, publisher of the *World Street Journal* started publishing share price *averages*. The Dow Jones average started with an average of twelve shares. Since 1928, the number of shares whose price average is calculated has been thirty. The main index in Japan, the Nikkei Dow 225, an index of 225 shares is based on average prices, not market capitalisation. Usually, the average is an arithmetic mean, but sometimes geometric mean is calculated. For example, the *Financial Times* Ordinary Share Index which began in 1935, calculated the geometric mean of the prices of 30 shares. Modern indices are based on taking the number of shares and multiplying it by the price. This gives accurate weight to the companies with the largest market capitalisation.

The majority of the stock market indices used in practice fall into two categories:

(i) equal-weighted index, and (ii) value-weighted index. The former is an index that reflects the simple arithmetic average of the sample shares at a certain time (year, month or day) concerning the base period. The latter reflects the aggregate market capitalisation of the sample shares at a certain time concerning the base period. Thus, what we had earlier referred to as price average can also be viewed as equal-weighted index and what we had referred simply to price index can be called value-weighted index.

Some analysts make a fine distinction even between the market capitalisation-weighted method and the non-capitalisation-weighted method. In this classification, the market capitalisation weighted method can be sub-classified into three types: **full market capitalisation method; free-float market capitalisation method and modified capitalisation method.** In the first, the number of shares outstanding multiplied by the market price of a company's share determines the scrip's weight in the index. The shares with higher market capitalisation would have greater weightage in this kind of index and will exercise greater influence on the overall index. In India, as has been mentioned earlier, the BSE Sensex follows this method. So does the Nifty. The free-float method does not consider a set constructed from among all shares in the market but from a smaller subset of shares. It considers only those shares that are freely available for buying in the market and excludes strategic investments in a company and shares locked in under employee stock-ownership plans. **The free-float method** recognises that investible market capitalisation may be lower than the total capitalisation. In this method, closely held companies would have a much lower weightage. The free-floating methodology is quite popular across the world. The third method under the market capitalisation method is the **modified capitalisation method.** This method recognises that if a simple market capitalisation method is followed, the largest stocks with huge market capitalisation would have a disproportionate weightage and would dominate the entire index. Hence, the modified capitalisation method seeks to limit the influence of the largest stocks in the index.

The two other methods for calculating stock price indices are the **price-weighted index** and **the equal-weighting method.** In the former, the price of each stock in the index is summed up which is then related to a base value. In the equal weighting method, each stock's percentage weight in the index is equal.

Let us now mention some of the major indices of share prices in India and abroad. In India, there are two main indices: the Bombay Stock Exchange (BSE) Sensitive index (Sensex) and the National Stock Exchange (NSE) Nifty. The BSE Sensex (sometimes called S&P BSE Sensex) was launched in 1986. It is a free-float market-weighted stock market index of 30 well-established and financially sound companies listed on the Bombay Stock Exchange. It uses large market capitalisation.

The other major share price index in India is the Nifty of the National Stock Exchange. The National Stock Exchange started its trading in capital market segment in 1994 and in a very quick time, its volume exceeded that of BSE. With collaboration with the Credit Rating Information Services of India Limited (CRISIL) and Standard & Poor, the S&P CNX Nifty was launched on July 8, 1996, comprising 50 scrips, which are selected based on low impact cost, high liquidity and market capitalisation. Both BSE and NSE use a weighted average method of weighting, where the weights are in proportion to its market capitalisation.

Other than these two major indices, some other indices are the **Economic Times Ordinary Share Price Index, Financial Express Ordinary Price Index, and the RBI Index of Security Prices.** Regional stock exchanges such as those in Ahmedabad, Chennai, Delhi and Kolkata have their indices.

We now state some major indices abroad. A very prominent index is the Dow Jones Industrial Average, which is for the New York Stock Exchange, the National Association of Securities Dealers Automatic Quotation (NASDAQ), the Nikkei in Japan, the Standard &Poor etc.

Check Your Progress 3

1) What is the difference between issue capital and paid-up capital? What do you understand by equity premium.?

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2) What are the functions performed by the secondary stock markets?

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11.7 LET US SUM UP

In this unit, light has been thrown on financial markets for long-term funds. We have discussed both debt and equity markets. We started by learning about the various participants in the capital market. We also came to know the important functions that capital markets perform. They perform important roles in providing information, enabling matching of sellers and buyers, and lowering search costs and transaction costs. The unit then went on to look at the comparative methods of raising finance through debt and equity. We studied this choice in the context of a representative business firm. This choice, which in general terms is called a firm's capital structure, is part of the firm's financial management.

We next turned to a discussion of debt markets and equity markets and took up their discussion one at a time, separately. We started with the debt markets looked at the instruments traded in that market, and discussed some theories regarding their pricing. We next studied equity markets, distinguishing between primary and secondary markets. We learnt about the way the issues are offered in the primary markets and we also looked at the functioning of secondary markets and learnt about their functions and uses. We studied in some detail the way to know if stock markets are advanced and mature in an economic system, or are they merely fledgling and poorly developed. We then learnt a bit about the volatility that characterises equity markets and how these are tackled. Finally, indices of share prices were discussed. We looked at the requirements of a good stock price index and came to know the various types of indices in use. The unit finally mentioned some of the important indices in India and abroad.

11.8 KEY WORDS

Coupon : The periodic interest paid on a bond. It equals the face value of the bond multiplied by the coupon rate of interest.

Coupon Rate : Annual coupon amount divided by the face value of the bond.

Equity Premium : When the issue price of an equity share is greater than the par value, the difference is called the equity premium.

Leverage : The ability of a firm to service its debt.

Market capitalisation : The sum of the market value of all the stocks included in the index.

Working capital : Current Assets minus Current Liabilities.

Par Value : Specified principal sum a bondholder is scheduled to receive when the bond matures.

11.9 SOME USEFUL BOOKS

Bhole, L.M. (2004), *Financial Institutions and Markets, 4th edition*, Tata McGraw-Hill, New Delhi.

Dothan, U. (1990), *Prices in Financial Markets*, Oxford University Press, New York

Fabozzi, F., Modigliani, F., and Jones, F. (2002) *Capital Markets, Institutions and Instruments 3rd edition*, Prentice-Hall, New Jersey.

11.10 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Mechanism for bringing together buyers and sellers of various types of financial assets, making information processing and dissemination easier, help to develop integration among several sectors of the economy.

2) Current ratio, quick or acid-test ratio; debt ratio and the debt-equity ratio.

Check Your Progress 2

1) value of the rate of interest that makes the present value of the cash stream received from owning the bond equal to the price of the bond.

2) $$A = \sum_{t=1}^{T} \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

Check Your Progress 3

1) The amount of capital that a company offers to investors is the issued capital. The part of capital subscribed by investors is called the paid-up capital. The par value is the face value of the share. When the issue price is greater than the par value, the difference is called the equity premium.

2) They provide information about the assets' fair price, as well as liquidity from their assets should they want to sell the assets, they reduce search costs, by making it easier for buyers and sellers of assets to find each other. They also lower transactions cost.

UNIT 12 BOND MARKETS

Structure

12.0 OBJECTIVES

After going through this unit, you will be able to:

- state the concept of the bond market;
- define bonds and categorize them according to their different types;
- evaluate the valuation of bonds across various scenarios, including regular coupon bonds, floating coupon bonds, and perpetual bonds;
- discuss bond yields such as Yield to Maturity, Yield to Call, and Current Yield; and
- elucidate and analyze credit rating,

12.1 INTRODUCTION

Suppose that a company needs ₹ 500 crore for some project and it does not want to approach a bank for a loan, then what are the alternatives available? One of the alternatives is the bond market. A bond market is a place where a company can raise loans by issuing a loan-taking certificate (technically known as a bond). Thus, an investor can purchase and sell bonds in a bond market. The present unit aims to discuss the functioning of a bond market and its product i.e., bonds. After reading this unit, you would be able to calculate the value of a bond and make the decision to sell or buy the bond. You will also be able to understand various terms used in the bond market e.g., Yield to Maturity, Yield to Call, Current Yield, Coupon rate, etc. Since bonds carry default and other types of risk, therefore, bonds with good ratings are preferred. The last section of this unit explains the credit rating.

12.2 MEANING OF BOND MARKET

Bonds are loans that investors make to borrowers' entities in exchange for interest payments (zero, fixed, or floating) made over a defined period.

The bond market serves as a way for the borrowers to raise capital, and for investors to earn interest income on their investments. Alternatively, a bond market (also known as a debt market, fixed-income market, or credit market) is the collective name given to all trades and issues of debt securities. The bond market broadly describes a marketplace where investors buy debt securities that are brought to the market by either governmental entities or corporations.

The bond market can be divided into two main categories: the **primary market** and the **secondary market**. In the primary market, new bonds are issued and sold to investors for the first time. In other words, the primary market is frequently referred to as the new issues market in which transactions strictly occur directly between the bond issuers and the bond buyers. In essence, the primary market yields the creation of brand-new debt securities that have not previously been offered to the public. In the **secondary market**, existing bonds are bought and sold among investors. The secondary market provides liquidity to investors, allowing them to sell their bonds before maturity if they need to raise cash or want to invest in other assets. In the secondary market, securities that have already been sold in the primary market are then bought and sold at later dates. Investors can purchase these bonds from a broker, who acts as an intermediary between the buying and selling parties. These secondary market issues may be packaged in the form of pension funds, mutual funds, and life insurance policies — among many other product structures.

The bond market is important to the broader economy because it helps to set interest rates for other financial products, such as mortgages and loans. It also plays a significant role in determining the cost of borrowing for corporations and governments, which in turn can impact their investment decisions and spending.

12.3 MEANING OF BONDS AND THEIR CLASSIFICATION

12.3.1 Meaning of Bonds

A bond is a long-term debt instrument in which the issuer of the bond is bound to pay a fixed or floating interest rate and repay the principal amount after a fixed period. Thus, a bond is a way of taking long-term funds.

A Real-Life Example: On March 14, 2023, Tata Capital Financial Services Limited (TCFSL) issued **bonds** worth ₹ 2,000 crore with a coupon rate of 8.3% p.a. and a maturity period of 2 years and 11 months. It means that TCFSL will receive ₹ 2,000 crore and pay fixed interest at 8.3% annually i.e., ₹ 16,600. These ₹ 2,000 crore will be repaid at the end of 2 years 11 months.

12.3.2 Characteristics of Bonds

1) **Long-term instrument**s – They are long-term debt instruments.

2) **Par value** – The par value is the stated face value of the bond. This value is mentioned on the bond certificate. It does not change.

3) **Carry coupon** – Except in the case of zero-coupon bonds, other bonds carry a fixed or floating interest rate known as a coupon rate. Such interest, generally, accrues annually, but it may be semi-annually or quarterly too. Moreover, interest is calculated on par or face value.

4) **Maturity period** – Bonds may have an indefinite maturity period, but generally they have a definite maturity period.

5) **Issued at par, premium, or discount** – They may be issued at par, premium or discount.

   a) If the issued price of a bond exceeds its face value, then it is said that the bond has been **issued at a premium**.

   b) If the issued price is equal to its face value, then it is said that the bond has been **issued at par.**

   c) If it is issued at a price less than its face value, then this is called a bond **issued at a discount**.

6) **Redemption** – Bonds may be redeemed at par or premium. If the redemption value of a bond exceeds its face value, then this is called redemption at a premium; if the redemption value of a bond is equal to its face value, then this is called redemption at par. Theoretically, redemption at a discount is also possible, but practically, it does not seem

to be feasible for an investor that he would get less than what he had invested.

12.3.3 Classification of Bonds

1) Bonds Based on Issuer

a) **Government bonds** – As the term implies these are those bonds which are issued by the Government (Central or State governments in the case of India). Indian bond market is majorly dominated by Government bonds. These bonds generally offer stable returns and are considered very safe because they are guaranteed by the Indian government. The interest rate on G-sec (a short form for Government securities or bonds) varies between 7% and 10%. In India, Government bonds may take the following forms:

i) Fixed-rate bonds (FRBs) – These bonds yield fixed interest payments throughout the life of the bonds.

ii) Floating rate bonds – These bonds yield variable interest payments.

iii) Sovereign Gold Bonds (SGBs) – These are the schemes in which an investor is allowed to invest in digital gold at an average price of gold without carrying gold in physical form. The Sovereign Gold Bond (SGB) Scheme 2022-23 is an example in which the interest rate is 2.5% p.a. on the nominal value of the bonds and they are redeemable on expiry of 8 years.

iv) Inflation Indexed Bonds – These bonds have a unique feature i.e., their prices and interest payments are adjusted according to the price level or inflation rate based on the consumer price index (CPI) or wholesale price index (WPI).

v) 7.75% GOI Savings Bonds – This government security was launched in 2018 to replace the 8% savings bond. The interest rate applicable here is 7.75%. The RBI stipulates that these bonds can be in the possession of individual(s) who aren't NRIs, minors, or are a Hindu undivided family.

b) **Corporate bonds** – These bonds are issued by corporations or companies. Since, Tata Capital Financial Services is a company in the above example, therefore, the bonds issued by it are corporate bonds. These bonds are generally considered to be riskier than government bonds. The creditworthiness of the issuing company is an important factor in determining the risk associated with the bond, as well as the interest rate offered to investors. Companies with high credit ratings are generally considered to be more creditworthy and may offer lower interest rates, while companies with lower credit ratings may offer higher interest rates to compensate for the higher risk of default. A few examples of corporate bonds are:

i) *LIC Housing Finance Limited Bonds* (coupon rate 7.95%, maturity date January 29, 2028, issue size ₹ 1477 crores)

ii) *Hindustan Petroleum Bonds* (coupon rate 7.74%, maturity date March 2028, issue size

iii) *Indian Railway Finance Corporation Limited Bonds* (issued date February 23, 2012, coupon rate 8.1%, maturity date February 23, 2027, face value ₹ 1,000).

c) **Municipal bonds** (**or muni**) – These bonds are issued by local authorities or local governments. For example, on February 20, 2023, *Indore Municipal Corporation* allotted bonds with a coupon rate of 8.25%, maturity date of February 20, 2026, and issue size ₹ 61crore.

d) **Foreign bonds** – These are bonds issued by a foreign government or foreign corporation.

2) Bonds Based on Coupon

a) **Zero coupon bonds (or pure discount bonds)** – These bonds do not carry any interest or coupon, but are issued at a discount and repaid/redeemed at par. For example, suppose that a bond has a face value of ₹ 100 and this is issued at a discount of ₹ 10, therefore, the issuer receives ₹ 90. Moreover, it will be repaid at ₹ 100. Thus, an investor by paying ₹ 90 will get ₹ 100 on the maturity of the bond. Thus, in the case of zero-coupon bonds, an investor earns capital return only not revenue return in the form of interest.

b) **Coupon bonds (or regular bonds)** – These bonds carry some interest. If they carry fixed interest, they are known as **fixed coupon bonds** and if they carry floating interest, they are known as **floating coupon bonds**. Thus, in the case of coupon bonds, an investor earns capital as well as revenue returns.

3) Bonds based on maturity

a) **Perpetual bonds** – These bonds have indefinite maturity.

b) **Non-perpetual bonds** – These bonds have a definite maturity period.

4) Bonds based on option

a) **Puttable bond (or retractable bonds)** – A put bond is a type of bond that gives the bondholder the right to sell the bond back to the issuer at a predetermined price and a specified date before the bond's maturity. The price at which the bond can be sold is called the put price, and it is usually higher than the bond's face value.

b) **Callable bond** – A callable bond is a type of bond that gives the issuer the right to redeem the bond before its maturity date. This means that the issuer can call or buy back the bond from the bondholder before the scheduled maturity date at a specified call price.

5) **Bonds based on convertibility**

a) **Convertible bonds** – These bonds can be converted into the shares of the issuer company depending upon the terms and conditions of the issue.

b) **Non-convertible bonds** – These bonds cannot be converted into the shares of the issuer company but they, generally, carry a higher interest rate than convertible bonds.

Check Your Progress 1

1) What do you mean by bonds? What are their characteristics?

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2) What are the types of bonds?

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12.4 VALUATION OF BONDS

Valuation of bonds means finding the **intrinsic value of a bond**. This intrinsic value is simply known as **the value of a bond**. The value of a bond is nothing but the present value of all the coupons and the redemption value of the bond. Therefore, the value of a bond can be calculated using the following formula:

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \cdots + \frac{C_n}{(1+i)^n}\right] + \frac{RV}{(1+i)^n}$$

$$V = \left[C_1 \times \frac{1}{(1+i)^1} + C_2 \times \frac{1}{(1+i)^2} + \cdots + C_n \times \frac{1}{(1+i)^n}\right] + RV \times \frac{1}{(1+i)^n}$$

$$V = \left[\sum_{k=1}^{n} C_k \times \frac{1}{(1+i)^k}\right] + \frac{RV}{(1+i)^n}$$

Where,

C_1, C_2, C_3, …C_n are coupons or interest payments over *n* years, *RV* is the redemption value, and ***i*** is the discount rate. Apart, from the expressions

$$\frac{1}{(1+i)^1}, \frac{1}{(1+i)^2}, \dots \dots, \frac{1}{(1+i)^n}$$

are known as the **present value factors** (PVFs) calculated for the first year, second year, third year, and so on. These present value factors can be calculated using

a) Present value tables,

b) log technique,

c) financial or scientific calculators,

d) MS-Excel

e) Various apps e.g., https://cleartax.in/s/present-value-calculator

Value of a Perpetual Bond

If a bond is a perpetual bond, then

$$V = \frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \cdots + \frac{C_n}{(1+i)^n} + \cdots \dots \dots \dots$$

If the coupons are constant i.e., $C_1 = C_2 = C_3 = \cdots = C$, then

$$V = \frac{C}{(1+i)^1} + \frac{C}{(1+i)^2} + \cdots + \frac{C}{(1+i)^n} + \cdots \dots \dots \dots$$

This is obvious that the above series is an infinite geometric progression with a common ratio

$$\frac{1}{1+i}$$

We know that if a is the first term and r is the common ratio of an infinite geometric progression, then its sum is given by the following formula:

$$S_\infty = \frac{a}{1-r}$$

Therefore,

$$\mathrm{V} = \frac{\mathrm{C}/(1+i)}{1-1/(1+i)}$$

$$V = \frac{C/(1+i)}{(1+i-1)/(1+i)}$$

$$V = \frac{C}{i}$$

The decision to buy bond:

1) **Overvalued bond** – If the market price of the bond exceeds its value, then it is called an overvalued bond, therefore, it should not be purchased.

2) **Undervalued bond** – If the market price of the bond is less than its value, then it is called an undervalued bond, therefore, it should be purchased.

The decision to sell a bond

Suppose you are the holder of a bond and you find that the market price of the bond exceeds its value, then you should sell the bond in the market. If the market price of the bond is less than its value, then it is better to hold it.

Example 12.1: Suppose that a corporation issues a bond having a face value ₹ 1,000 redeemable at a premium of 10% at the end of 3 years. If the discount rate is 12% and the coupon rate is 8%, then calculate the value of this bond. Moreover, would you buy this bond if the market price of the bond is ₹ 1,050 or ₹ 800?

Solution

Step 1: Summarise the information

Face value = ₹ 1,000

Coupon rate = 8%

Coupon = Coupon rate × face value = 8% × ₹ 1,000 = ₹ 80

Discount rate = i = 12% = 0.12

Maturity of bond = n = 3 years

Redemption value = RV = Face value + premium of 10% = 1,000 + 10% × 1,000 = ₹ 1100

Step 2: Apply the following formula

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \frac{C_3}{(1+i)^3} + \cdots + \frac{C_n}{(1+i)^n}\right] + \frac{RV}{(1+i)^n}$$

$$= \left[80 \times \frac{1}{(1+0.12)^1} + 80 \times \frac{1}{(1+0.12)^2} + 80 \times \frac{1}{(1+0.12)^3} + \right] + 1100 \times \frac{1}{(1+0.12)^3}$$

$$= \left[80 \times \frac{1}{(1.12)^1} + 80 \times \frac{1}{(1.12)^2} + 80 \times \frac{1}{(1.12)^3} + \right] + 1{,}100 \times \frac{1}{(1+0.12)^3}$$

$$= [80 \times 0.8929 + 80 \times 0.7972 + 80 \times 0.7118 +] + 1{,}100 \times 0.7118$$

$$= 71.43 + 63.76 + 56.94 + 782.98$$

$$= 975.13$$

Thus, the value of the bond is ₹ 975.13.

The decision to buy bond:

1) If the market price is ₹ 1,050, then it exceeds the value ₹ 975.13, therefore, the bond is overvalued and should not be bought.

2) If the market price is ₹ 800, then it is less than the value of ₹ 975.13, therefore, the bond is undervalued and should be bought.

Example 12.2: What is the value of a perpetual bond with a coupon of 8% and face value ₹ 1000? Assume a discount rate of 10%.

Solution

Coupon = C = 8% × ₹ 1,000 = ₹ 80

Discount rate = *i* = 10% = 0.10

Therefore, the value of the bond is

$$V = \frac{C}{i} = \frac{80}{0.10} = ₹\,800$$

Check Your Progress 2

1) How are bonds valued?

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2) A bond with a face value of ₹ 5,000 matures at the end of 5 years, the rate of interest on the bond being 15% p.a. paid annually. Find the value of the bond to yield a return of 10% p.a. Further, if the bond is trading at ₹ 6,000, then is it worth buying it? Assume that the bond will be redeemed at par. (**Hint: Find the present value of all the benefits from the bond. Refer to example 12.1)**

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3) Will your decision be different if the bond is traded at ₹ 5,500? (**Hint: Refer to example 12.1**)

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12.5 BOND YIELDS

12.5.1 Yield to Maturity (YTM)

From the previous section, we have the formula to calculate the value of a bond i.e.,

$$V = \left[\sum_{k=1}^{n} C_k \times \frac{1}{(1+i)^k}\right] + \frac{RV}{(1+i)^n}$$

So, if we have the values of C_k for k = 1, 2, 3, …., *n*, *RV* and discount rate *i*, then we can easily calculate the value of the bond. Now suppose that an investor already knows the value of a bond in terms of its price, say, P and he is interested in finding a discount rate *i*, such that the above equation is satisfied, then this discount rate is called Yield to Maturity (YTM). So, YTM means the return on a bond (or discount rate) which makes the present value of all the coupons and redemption value equal to the price of bond P provided that the bond is held till its maturity. This is the return on investment which investors talk about when they invest in bonds. Thus, YTM means that *i* which satisfies the following equation:

$$P = \left[\sum_{k=1}^{n} C_k \times \frac{1}{(1+YTM)^k}\right] + \frac{RV}{(1+YTM)^n}$$

Although we have a **trial and error approach (**a very time-consuming approach**)** to calculate YTM, we have an approximation formula given below to calculate YTM if coupons are constant.

$$\text{Approximate YTM} = \frac{C + (F-P)/n}{(F+P)/2}$$

Where C = Constant coupon, F = Face value, P = Price of the bond

Example 12.3: Suppose that there is a bond with a face value of ₹ 1,000 and a coupon rate of 10%. If the maturity is 5 years and the price of the bond is ₹ 800, then calculate the approximate YTM. Interpret your result.

Solution:

$$\text{Approximate YTM} = \frac{C + (F - P)/n}{(F + P)/2}$$

$$= \frac{100 + (1000 - 800)/5}{(1000 + 800)/2}$$

$$= \frac{100 + 40}{900}$$

$$= \frac{140}{900}$$

$$= 0.1555 \text{ or } 15.55\%$$

Interpretation: It means that if ₹ 800 is invested in the bond held for 5 years, then it will give us a return of 15.55%.

12.5.2 Yield to Call (YTC)

In the case of a callable bond, the issuer can call or buy back the bond from the bondholder before the scheduled maturity date at a specified call price. Thus, redemption can be done before maturity. In such a case, the rate of return on the bond till the call date is called Yield to Call. Thus, YTC is that *i* which satisfies the following equation:

$$\mathrm{P} = \left[\sum_{\mathrm{k}=1}^{\mathrm{s}} \mathrm{C_k} \times \frac{1}{(1 + \mathrm{YTC})^{\mathrm{k}}}\right] + \frac{\text{Call price}}{(1 + \mathrm{YTC})^{\mathrm{s}}}$$

where,

s = Period up to call

12.5.3 Current Yield

The current yield is the annual interest payment divided by the current price of the bond. Thus, the current yield formula is

$$\frac{\text{Annual interest or coupon}}{\text{Current price of bond}}$$

Therefore, if a company's bonds with a 10% coupon were currently selling at ₹ 980, the bond's current yield would be 10.20% (₹100/₹980). It means by investing ₹ 1 now, an investor will earn ₹ 0.102 as interest at the end of 1 year. Current yield focuses on interest only not the redemption value and this is the reason we say that current yield is not an appropriate measure of yield.

Check Your Progress 3

1) What is **Yield to Maturity (**YTM)?

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2) Calculate the yield to maturity from the following information. Coupon rate = 7%, face value = ₹ 500, price of bond = ₹ 350, maturity period = 10 years. **(Hint: Use the approximation formula)**.

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3) What are Yield to Call and Current Yield?

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12.6 CREDIT RATING

Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full. The agency performing credit rating is known as **a credit rating agency**. Credit rating agencies, assign credit ratings based on the borrower's financial history, including factors such as income, assets, liabilities, repayment track record, and other relevant financial indicators. The credit rating agencies use a letter-based rating system to indicate the creditworthiness of borrowers, with AAA being the highest rating and D being the lowest. Credit ratings are important because they help lenders and investors assess the risk associated with lending money to a borrower. A high credit rating means that the borrower is less likely to default on their loans, which can lead to lower interest rates and better loan terms. On the other hand, a low credit rating can make it more difficult to obtain credit and can result in higher interest rates and less favourable loan terms.

Worldwide, perhaps, the most popular credit rating agencies are **Moody's Investors Service (Moody's) and Standard & Poor's (S&P)**. In India, as of now, there are seven registered credit rating agencies. Their names are as follows:

1) Credit Rating Information Services of India Limited (CRISIL)
2) Investment Information and Credit Rating Agency of India (ICRA) Limited
3) Credit Analysis and Research (CARE) Limited
4) Acuite Ratings & Research,
5) Brickwork Ratings India Private Limited
6) India Ratings and Research Pvt. Ltd.
7) Infometrics Valuation and Rating Pvt. Ltd.

CRISIL is the oldest credit rating agency in India followed by ICRA and CARE. Let us see briefly their credit rating notations and explanations.

1) Credit Rating Information Services of India Limited (CRISIL)

Credit Rating Information Services of India Limited (CRISIL), established in 1987, is one of the oldest credit rating agencies in India. It operates in various countries like the UK, USA, Poland, China, Hong Kong, and Argentina, apart from India. It evaluates the creditworthiness of commercial entities based on their strengths, market reputation, market share and board. Its ratings range from CRISIL AAA to CRISIL D which are given below:

Table 12.1: Different categories of credit ratings given by CRISIL

Rating	Explanation	Credit Risk	Degree of safety
CRISIL AAA	Securities with this rating are considered to have the highest degree of safety regarding timely servicing of financial obligations. Such securities carry the lowest credit risk.	Lowest risk	Highest degree of safety
CRISIL AA	Securities with this rating are considered to have a high degree of safety	Very low risk	The high degree of safety

	regarding timely servicing of financial obligations. Such securities carry very low credit risk.		
CRISIL A	Securities with this rating are considered to have an adequate degree of safety regarding the timely servicing of financial obligations. Such securities carry low credit risk.	Low risk	Adequate degree of safety
CRISIL BBB	Securities with this rating are considered to have a moderate degree of safety regarding timely servicing of financial obligations. Such securities carry moderate credit risk.	Moderate risk	A moderate degree of safety
CRISIL BB	Securities with this rating are considered to have a moderate risk of default regarding timely servicing of financial obligations.	Moderate risk of default	-
CRISIL B	Securities with this rating are considered to have a high risk of default regarding the timely servicing of financial obligations.	High risk of default	-
CRISIL C	Securities with this rating are considered to have a very high risk of default regarding the timely servicing of financial obligations.	Very high risk of default	-
CRISIL D	Securities with this rating are in default or are expected to be in default soon.	Worst risk	-

Source: *https://www.crisilratings.com/en/home/our-business/ratings/credit-ratings-scale.html*

2) Investment Information and Credit Rating Agency of India (ICRA) Limited

The Investment Information and Credit Rating Agency of India, established in 1991, uses a rating system to give comprehensive credit ratings to corporates. ICRA is known for assigning corporate governance ratings, mutual funds ratings, structured finance ratings, performance ratings, etc. Credit ratings of ICRA range from ICRA AAA to ICRA D. Their explanations are the same as in CRISIL.

3) Credit Analysis and Research (CARE) Limited

Credit Analysis and Research Limited (CARE) was established in 1993. CARE rates bank loan ratings- short-term and long-term debt instruments. Credit ratings of CARE range from CARE AAA to CARE D. Their explanations are the same as in CRISIL.

Check Your Progress 4

1) What is a credit rating? How is credit rating useful for an investor?

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2) What are the credit rating codes of CRISIL, ICRA, and CARE?

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12.7 BOND MARKET IN INDIA

12.7.1 Bond Market — A Part of The Debt Market in India

The debt market, also known as the 'fixed income market,' includes instruments that offer fixed returns, with the bond market being a key component. In countries like India, debt markets have become a vital source of funds, serving as an alternative to traditional banking channels for financing. The significance of debt markets in an economy is underscored by their role in:

a) Efficiently mobilizing and allocating resources.

b) Funding the government's developmental activities.

c) Transmitting signals for the implementation of monetary policy.

d) Managing liquidity in line with short-term and long-term objectives.

e) Providing a credible alternative to banks as funding sources.

12.7.2 Regulators

As we have discussed in Unit 10 and Unit 11, Debt market transactions, including issuances and investments, fall under the jurisdiction of the following regulators:

a) **RBI (Reserve Bank of India):** As the 'money market' regulator, the Reserve Bank of India regulates and facilitates government bonds and other securities on behalf of the government.

b) **SEBI (Securities Exchange Board of India):** As the 'securities market regulator,' the Securities and Exchange Board of India regulates corporate and municipal bonds issued by entities that are listed or intend to be listed.

However, for money market instruments such as Commercial Paper, if the issuer intends to list them, the disclosures must comply with SEBI requirements.

12.7.3 The Growth of Indian Bond Market

India's bond market is currently valued at US$2.59 trillion, reflecting a 12.3% annual growth in INR terms from the previous year. The corporate bond segment, worth US$567 billion, also saw a year-over-year growth of 12.6%. Despite this, the debt market is predominantly composed of government securities (G-secs). Specifically, the outstanding stock of government securities amounts to ₹1,68,85,262 crores (US$2.02 trillion), while corporate bonds stand at ₹47,28,935 crores (US$567 billion). Over the past decade (from 2013-2023), India's bond market has grown at an impressive annual rate of 12.6% in INR terms, underscoring its expanding role in the country's financial ecosystem. The inclusion of Indian bonds in global indices by JP Morgan and Bloomberg has further highlighted Indian fixed income on the global stage. S&P Global Ratings has upgraded India's credit rating outlook from 'stable' to 'positive,' citing policy consistency, significant economic reforms, and substantial infrastructure investments as key drivers of sustained long-term growth. With India's GDP projected to reach US$5 trillion in the coming years, much of this growth will be financed through credit provided to corporations via the bond markets. This trend is evident in Western economies, where corporate bond markets constitute a significant percentage of GDP (up to 50-65% in some cases). As India transitions from a developing to a developed economy, the sophistication and depth of its financial markets will be crucial for economic advancement. Households transferring their substantial savings to investments, particularly into the safer asset class of corporate bonds, are expected to drive exponential growth in corporate bonds and support India's credit expansion.

Source: https://www.indiabonds.com/bonduni/blogs/size-of-the-indian-bond-market-is-us-2-59-trillion/

12.7.4 Classification of the Debt Instruments

In India, the classification (based on the issuer) of the debt or bond instruments is given below:

<table>
<tr><th colspan="2">Market segment</th><th>Issuer</th><th>Instruments</th></tr>
<tr><td colspan="2">Government Bonds or
Government securities</td><td>Central, State Governments</td><td>Zero coupon bonds, Coupon bearing bonds, Treasury Bills, STRIPS, State Development Loans, Gold Bonds, Sovereign Green Bonds</td></tr>
<tr><td rowspan="5">Corporate Bonds</td><td rowspan="2">Public sector bonds</td><td>Government Agencies /Statutory Bodies</td><td>Government Guaranteed Bonds and Debentures,
Municipal Bonds</td></tr>
<tr><td>Public Sector Units (PSU)</td><td>Bonds, commercial papers</td></tr>
<tr><td rowspan="3">Private sector bonds</td><td>Corporates</td><td>Bonds, commercial papers</td></tr>
<tr><td>Banks</td><td>Certificate of Deposits, Bonds, Structured instruments and Perpetual Bonds</td></tr>
<tr><td>Financial institutions</td><td>Certificate of Deposits, Bonds, Structured
Instruments, Commercial Paper</td></tr>
</table>

Let us discuss these debt instruments briefly:

Government Securities: G-Secs are issued by the Reserve Bank of India on behalf of the Government of India. Normally the dated government securities have a period of 1 year to 30 years. These are sovereign instruments generally bearing a fixed interest rate with interests payable semi-annually and principle as per schedule. For shorter term, RBI issues Treasury Bills which are discounted papers. At present T-Bills are issued for 91 days, 182 days & 364 days. G-Secs provide risk free (credit risk) return to investors.

STRIPS is the acronym for *Separate Trading of Registered Interest and Principal Securities*. Stripping is the process of separating a standard coupon-bearing bond into its individual coupon and principal components. For example, a 5-year coupon bearing (Government) bond (with half yearly coupon payment) can be stripped into 10 coupons and one principal instrument, all of which thenceforth would become zero coupon bonds.

Sovereign Green Bonds : The budget division of the ministry submits projects approved by GFWC for bond issuance through the Reserve Bank of India (RBI). These bonds, known as sovereign green bonds, are issued by the central bank to finance green initiatives. The Ministry of Finance maintains a green register containing bond details to monitor these bonds. Notably, the ministry releases an annual report on these projects, detailing bond issuances, approved projects, allocated expenditures, current statuses, and potential impacts on reducing carbon intensity.

Certificate of Deposit (CD) : As discussed in Unit 10, CDs are negotiable money market instruments issued in demat form or as a Usance Promissory Notes. CDs issued by banks should have a maturity of not less than seven days and not more than one year. Financial Institutions are allowed to issue CDs for a period between 1 year and up to 3 years. CDs normally give a higher return than Bank term deposit. CDs are rated by approved rating agencies (e.g. CARE, ICRA, CRISIL, FITCH) which considerably enhances their tradability in secondary market. CDs are issued in denominations of Rs. 1 Lac and in the multiples of Rs. 1 Lac thereafter.

Commercial Papers : As we discussed in Unit 10, a CP is a short-term security (7 days to 365 days) issued by a corporate entity (other than a bank), at a discount to the face value. One can invest in CPs starting from a minimum of 5 lacs (face value) and multiples thereof. CPs are rated by approved rating agencies (e.g. CARE, ICRA, CRISIL, FITCH). CPs normally give a higher return than fixed deposits & CDs. We deal in investment grade CPs only. CPs can be traded in the secondary market, depending upon demand. An element of credit risk is attached to CPs.

12.8 LET US SUM UP

Bond is a way of raising long-term loans. A bond is traded in a market known as a bond market which is classified as a primary market and a secondary market. The primary market deals in new bonds while the secondary market deals in second-hand bonds. The market forces of demand and supply determine the market price (or just price) of the bond, but its intrinsic value is determined by calculating the present value of all the coupons and redemption value. There are three types of bond yields: Yield to Maturity

(YTM), Yield to Call (YTC), and Current Yield. These yields are generally quoted by the issuers to attract investors. Bonds are of various types e.g., government bonds, corporate bonds, etc. Bonds carry default risk too; therefore, credit rating agencies rate the bonds or issuer companies/governments so that investors can know the creditworthiness of the issuer.

12.9 KEY WORDS

Bonds	:	Bonds are long-term debt instruments.
Coupon	:	It is the interest amount on a bond.
Bond market	:	A place where bonds are traded.
Value of bond	:	The value of a bond means the intrinsic value of a bond. It is calculated as the present value of all the coupons and the redemption value of it.
Price of bond	:	It is the market price of a bond which is determined by the rules of demand for and supply of bond.
Yield to Maturity (YTM)	:	It is the rate of return on a bond if the bond is held till its maturity.
Yield to Call (YTC)	:	In the case of a callable bond, it is the rate of return on a bond if the bond is held till the time of the call.
Current yield	:	It is a rate of return in terms of interest per ₹ 1 of investment.
Credit rating	:	Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full.
Credit rating agency	:	It is the agency which performs credit rating. CRISIL, CARE, and ICRA are examples of credit rating agencies in India.

12.10 SOME USEFUL BOOKS

Brigham, E., & Ehrhardt, M. (2017). *Financial Management* (12th ed.). Thomson South Western.

Horne, J. V., & Wachowicz, J. (2008). *Fundamentals of Financial Management* (13th ed.). Prentices Hall.

Kishore, R. M. (2010). *Strategic Financial Management* (2nd ed.). Taxmann Publications Pvt. Ltd.

Pandey, I. M. (2009). *Financial Management* (9th ed.). Vikas Publishing House Pvt. Ltd.

Luenberger, D. G. (1998). *Investment Science*. Oxford University Press.

12.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A bond is a long-term debt instrument in which the issuer of the bond is bound to pay a fixed or floating interest rate and repay the principal amount after a fixed period. Thus, a bond is a way of taking long-term funds.

2) **Bonds Based on Issuer:** Government bonds, Corporate bonds, Municipal bonds, Foreign bonds. **Bonds Based on Coupon:** Zero coupon bonds, Coupon bonds. **Bonds based on maturity:** Perpetual bonds, non-perpetual bonds**. Bonds based on option:** Puttable bond, Callable bond**. Bonds based on convertibility**: Convertible bonds, non-convertible bonds.

Check Your Progress 2

1) Refer to Section 12.4

2) A bond with a face value of ₹ 5,000 matures at the end of 5 years, the rate of interest on the bond being 15% p.a. paid annually. Find the value of the bond so as to yield a return of 10% p.a. Further, if the bond is trading at ₹ 6,000, then is it worth to buy it? Assume that the bond will be redeemed at par.

Step 1: Summarise the information

Face value = ₹ 5,000

Coupon rate = 8%

Coupon = Coupon rate × face value = 15% × ₹ 5,000 = ₹ 750

Discount rate = i = 10% = 0.10

Maturity of bond = n = 5 years

Redemption value = RV = Face value = ₹ 5,000

Step 2: Apply the following formula

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \frac{C_3}{(1+i)^3} + \cdots + \frac{C_n}{(1+i)^n}\right] + \frac{RV}{(1+i)^n}$$

$$= \left[\frac{750}{(1+0.10)^1} + \frac{750}{(1+0.10)^2} + \frac{750}{(1+0.10)^3} + \frac{750}{(1+0.10)^4} + \frac{750}{(1+0.10)^5}\right] + \frac{5{,}000}{(1+0.10)^5}$$

$$= \left[\frac{750}{(1.10)^1} + \frac{750}{(1.10)^2} + \frac{750}{(1.10)^3} + \frac{750}{(1.10)^4} + \frac{750}{(1.10)^5}\right] + \frac{5,000}{(1.10)^5}$$

$$= [750 \times 0.9090 + 750 \times 0.8264 + 750 \times 0.7513 + 750 \times 0.6830 + 750 \times 0.6209 +] \\ + 5,000 \times 0.6209$$

$$= [681.8182 + 619.8347 + 563.4861 + 512.2601 + 465.691] + 3104.607$$

$$= 5947.697$$

Since the bond is being traded at ₹ 6,000 and the intrinsic value is ₹ 5947.697, therefore, the bond is undervalued and it is worth to buy it.

Check Your Progress 3

1) Refer to Sub-section 12.5.1

2) Calculate the yield to maturity from the following information. Coupon rate = 7%, face value = ₹ 500, price of bond = ₹ 350, maturity period = 10 years.

$$\text{Approximate YTM} = \frac{C + (F - P)/n}{(F + P)/2}$$

$$\text{Approximate YTM} = \frac{35 + (500 - 350)/10}{(500 + 350)/2}$$

$$= \frac{35 + 15}{425}$$

$$= \frac{50}{425}$$

$$= 0.1176$$

$$= 11.76\%$$

Note: The above formula is approximation only. If you are interested to find the exact YTM, then the trial-and-error method will give us 12.396% as YTM. In order to understand the trial-and-error method, the students are suggested to read to calculate IRR by referring any book on Financial Management.

3) Refer to Sub-section 12.5.2 and Sub-section 12.5.3

Check Your Progress 4

1) Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full.

2) Refer to Table 12.1

UNIT 13 DERIVATIVES

Structure

13.0 OBJECTIVES

After going through this unit, you will be able to:

- state the meaning of derivatives with their characteristics;
- explain the main types of derivatives;
- discuss the pros and cons of derivatives; and
- distinguish between futures and forward contract.

13.1 INTRODUCTION

In Unit 10 and 11, you have understood various types of financial assets e.g., equity shares, stock, debentures, fixed deposits, and so on. All these assets share a common trait that their values are not linked to the value of some other asset. However, there are some unique financial assets known as derivatives which derive their values from some other assets known as underlying assets. These underlying assets may be shares, stocks, debentures, currencies, etc. Derivatives can be used to manage various types of risks, such as price, interest rate, and credit risk, as well as to speculate on future price movements of the underlying asset. In this unit, we shall discuss various aspects of derivatives.

13.2 MEANING OF DERIVATIVES

In finance, a derivative is a financial instrument that derives its value from an underlying asset, such as a stock, bond, currency, or commodity. The value of a derivative is based on the changes in the value of the underlying asset. Simply put, if an asset, say A, is valued based on another asset, say B, which is linked to it, then A is called a derivative and B is called an underlying asset. The most common types of derivatives include *options, futures, forwards, and swaps*. Derivatives play an important role in financial markets, allowing investors and companies to manage risk and enhance returns, but they can also be complex and carry significant risk if not used properly.

The Securities Laws (Second Amendment) Act, 1999, has officially included derivatives in the Securities definition. According to the Securities Contracts (Regulations) Act, a derivative comprises securities derived from debt instruments, shares, loans, risk instruments, or contracts for differences. It also includes contracts deriving their value from the prices or index of underlying securities.

13.3 CHARACTERISTICS OF DERIVATIVES

The characteristics of derivatives are as follows:

a) A derivative derives its value from an underlying asset like a share or stock of a company. This feature makes it a unique financial asset.

b) Derivatives are traded globally and are very popular in developed economies.

c) Common derivatives include futures contracts, forward contracts, options, and swaps.

d) Most derivatives are not traded on exchanges and are used by institutions to hedge risk or speculate on price changes in the underlying asset.

e) Exchange-traded derivatives like futures or stock options are standardized and eliminate or reduce many of the risks of over-the-counter derivatives.

f) Derivatives are usually leveraged instruments, which increases their potential risks and rewards.

Advantages of Derivatives

a) **Lock in prices:** Derivatives help to protect the securities against fluctuations in prices. Thus, it helps to reduce market uncertainties. Companies and investors can hedge against price fluctuations in commodities, currencies, or securities, ensuring they are not adversely affected by market volatility. It also helps in financial planning and budgeting as the market uncertainty is reduced.

b) **Hedge against risk**: Derivatives are widely used to hedge against various types of risks, providing a safeguard for investors and businesses. Businesses involved in the production or consumption of commodities can use futures or options to lock in prices, protecting themselves from adverse price movements. Companies with debt or fixed-income investments can use interest rate swaps or futures to hedge against fluctuations in interest rates.

c) **Derivatives can be leveraged:** They allow investors to gain exposure to a larger position than what would be possible by investing directly in the underlying asset with the same amount of capital. Leverage can amplify returns on investment because a small change in the price of the underlying asset can lead to a larger change in the value of the derivative.

d) **Useful in making a diversified portfolio:** Derivatives can provide exposure to a wide range of asset classes, including commodities, currencies, interest rates, and equities, thus providing scope for diversification. Investors can also use derivatives to gain exposure to emerging markets, which might be otherwise inaccessible due to regulations or high entry costs.

Limitations of derivatives

a) **Difficult to value them:** Derivatives often require complex mathematical models to determine their fair value. For example, options pricing commonly uses the Black-Scholes model or binomial models, which involve advanced mathematics and assumptions about market behaviour. We will discuss Black-Scholes model and binomial pricing models in Unit 21.

b) **Subject to counterparty default:** If a counterparty defaults, the non-defaulting party may incur significant financial losses as the value of a derivative can be heavily dependent on the creditworthiness of the counterparty. A decline in the counterparty's credit rating can decrease the value of the derivative.

c) **Complex to understand:** There are numerous types of derivatives, such as options, futures, forwards, swaps, and more complex structures like collateralized debt obligations (CDOs) and credit default swaps (CDS). Each type has its own set of rules, characteristics, and uses. Moreover, pricing of derivatives also involves complex procedure.

d) **Sensitive to demand and supply factors**: The prices of derivatives can be highly volatile due to changes in demand and supply in the underlying asset markets. Moreover, high levels of speculative trading can amplify price movements and thus make derivatives volatile.

13.4 A BRIEF HISTORY OF DERIVATIVES IN INDIA

The historical roots of derivatives extend further back than commonly acknowledged, with some texts even suggesting their existence in incidents dating back to the Mahabharata. Traces of derivative contracts can be found in historical occurrences predating the era of Jesus Christ. Nevertheless, the formalization of modern derivative contracts is often attributed to the necessity for farmers to safeguard against potential declines in crop prices caused by delayed monsoons or overproduction.

India has a longstanding association with derivatives, particularly in the commodity market. Organized trading in cotton, dating back to 1875 with the establishment of the Cotton Trade Association, marked the initiation of the commodity derivative market in the country. Over the years, contracts for various other commodities have been introduced. In June 2000, exchange-traded financial derivatives were introduced at major stock exchanges in India, namely NSE and BSE. The National Commodity & Derivatives Exchange Limited (NCDEX) commenced operations in December 2003, providing a platform for commodities trading.

The derivatives market in India has witnessed substantial growth, especially at NSE. As of April 13, 2005, Stock Futures constitute the most actively traded contracts on the NSE, contributing to approximately 55% of the total turnover of derivatives.

Derivative trading in India can occur either on a distinct Derivative Exchange or within a designated segment of an existing Stock Exchange. These Derivative Exchanges/Segments operate as Self-Regulatory Organizations (SROs), with SEBI serving as the overseeing regulator. The clearing and settlement of all derivative trades on these platforms are facilitated through independent Clearing Corporations/Houses.

The introduction of derivative products in India occurred in stages, beginning with Index Futures Contracts in June 2000. Subsequently, Index Options and Stock Options were introduced in June and July 2001, respectively, followed by Stock Futures in November 2001. Derivatives trading in sectoral indices gained approval in December 2002. In June 2003, Interest Rate Futures on a notional bond and T-bill priced off ZCYC were introduced. Furthermore, exchange-traded interest rate futures on a notional bond priced off a basket of Government Securities received approval for trading in January 2004.

Check Your Progress 1

1) What do you mean by derivatives? Why are they called derivatives?

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2) What are the characteristics of derivatives?

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3) Discuss the advantages and disadvantages of derivatives.

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13.5 TYPES OF DERVIATIVES

Derivatives can be categorized based on various criteria, such as the nature of the underlying asset, trading venue, and contract structure. Although there are many different forms of derivatives, the following are some of the most popular ones in finance are:

1) **Futures:** Futures contracts are agreements to buy or sell an underlying asset at a predetermined price and time in the future. They are commonly used to hedge against price fluctuations and are traded on organized exchanges.

2) **Forwards:** Forwards are similar to futures, but they are not traded on exchanges. Instead, they are customized contracts between two parties to buy or sell an underlying asset at a predetermined price and time in the future.

3) **Options:** Options are contracts that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a predetermined price and time. They can be used for hedging or speculative purposes, and are traded on both exchanges and over-the-counter markets.

4) **Swaps:** Swaps are agreements to exchange cash flows based on different financial instruments, such as interest rates, currencies, or commodities. They are commonly used to manage risk and are traded over the counter. The two most popular swaps are interest rate swaps and credit default swaps (CDS).

5) **Credit derivatives:** Credit derivatives are contracts that allow investors to manage credit risk by transferring it to another party. They include credit default swaps, which allow investors to protect themselves against

default on a debt obligation, and collateralized debt obligations (CDOs), which allow investors to pool together different types of debt securities.

6) **Equity derivatives:** Equity derivatives are contracts that allow investors to gain exposure to changes in the price of a stock or stock index. They include options, futures, and swaps on individual stocks or broad market indices.

7) **Commodity derivatives:** Commodity derivatives are contracts that allow investors to gain exposure to changes in the price of a commodity, such as gold, oil, or wheat. They include futures, options, and swaps on various commodities.

13.6 FUTURES AND FORWARDS

13.6.1 Meanings of Futures and Forwards

1) **Forwards** – A forward contract is a financial agreement between two parties where they agree to buy or sell an asset at a predetermined price at a future date. The parties involved in a forward contract can be individuals, corporations, or financial institutions. Forward contracts are used to hedge against price fluctuations in an asset, such as commodities, currencies, stocks, or bonds.

   Example 13.1: If a company wants to buy a certain commodity at a future date, it can enter into a forward contract with a supplier to buy the commodity at a fixed price, thus avoiding the risk of price volatility. Similarly, if an investor expects a certain stock to appreciate, they can enter into a forward contract to buy the stock at a predetermined price in the future.

2) **Futures** – Futures contracts are financial instruments that allow parties to buy or sell an asset, such as commodities, currencies, or financial instruments, at a predetermined price and date in the future. The price of the underlying asset is fixed at the time the futures contract is created, and the transaction is legally binding.

   Futures contracts are often used by traders and investors to hedge against potential price fluctuations or to speculate on the direction of market movements. For example, a farmer might sell a futures contract on a commodity like wheat to lock in a price for a future delivery, thereby reducing the risk of a price drop. Similarly, a speculator might buy a futures contract on a stock index if they believe that the market will rise in the future.

   We will discuss more about Futures and Forwards in Unit 21-Pricing of Derivatives.

13.6.2 Differences between Futures and Forwards

Futures and Forwards can be differentiated as follows:

Table 13.1: Differences between Futures and Forwards

S.No.	Forward contracts	Futures contracts
1	The contract price is not publicly disclosed	The contract price is transparent.
2	The contract is exposed to default risk by the counterparty.	The contract has effective safeguards against defaults.
3	Each contract is unique in terms of size, expiration date and asset type.	The contracts are standardized in terms of size, expiration date and all other futures.
4	Forward contracts suffer from the problem of liquidity.	There is no problem of liquidity.
5	Settlement of the contract is done by delivery of the asset on the expiration date.	Settlement is done on a cash basis.
6	They are traded on a personal basis or the telephone or otherwise.	They are traded in a competitive market.
7	They are traded in an over-the-counter market.	They are traded on organised exchanges with a designated physical location.
8	The costs of forward contracts are based on the bid-ask spread.	They involve brokerage frees.
9	Forward contracts are not subject to marking to market.	They are subject to marking to market.
10	Credit risk is borne by each party.	Credit risk is not borne by each party.

13.7 OPTIONS

An option is an agreement between two parties (say A and B) in which a party, say, A, has the right to buy or sell a particular asset at a predetermined price within a specified time frame. Moreover, if A does not want to execute his right, then he cannot be forced to do so by B. That is why the agreement is called option not obligation. An option helps in managing the risk of ups and downs of the price of an asset in which A is interested. Making a strategy to reduce or manage risk is called hedging. So, an option is a hedging tool.

Now a question arises i.e., **when should A use his right (technically known as exercising option)?** Before answering this question, we should understand two main types of options i.e.

1) Call Option

2) Put Option

13.7.1 Call Option

Suppose that the current price of a share of a company is ₹ 500. There is a good chance that the price will go up in the next 4 months, but a fall is also probable in the price. You are in a dilemma to buy the share or not. Luckily, you find a person, say, Vikas, who agrees to give you an option to buy the share from him at ₹ 530 after 4 months irrespective of the actual price of the share in the market. For this, Vikas will charge a cost of ₹ 20 as purchase cost. Common sense says that if the actual price of the share goes up, say, ₹ 580, then it will be profitable for you to purchase the share from Vikas at ₹ 530 and sell in the market at ₹ 580. It will create a profit of ₹ 30 after charging the cost of ₹ 20. Of course, it will be a loss-making situation for Vikas. Now, what if the price does not reach the level which creates profit for you? In that case, you cannot be forced to buy the share from him because it is an option, not an obligation. Thus, by paying an amount of ₹ 20 only, you can avail the chance or opportunity of profit with zero risk or you can hedge your risk.

In finance, such an agreement is called a **call option**, the share of the company is called **underlying asset**, the price demanded by Vikas is called the **exercise price or strike price**, and the difference between the exercise price at expiry and the price of the underlying asset i.e., ₹ 50 is called exercise value (or intrinsic value of the call option), the cost of ₹ 20 charged by him is called **option premium**, the profit of ₹ 30 earned is called net **payoff**, and Vikas is called **option writer** and you are an **option buyer**. Since this call option derives its value because of a share linked to it, therefore, it is a derivative also. Thus, we can define a call option as follows:

A **call option** is a type of derivative in which the buyer gets a right to **buy** an underlying at an agreed price (known as exercise price or strike price) when the buy is profitable to him at the time of expiry of the option. Since this derivative is an option, therefore, he cannot be forced by the option writer to buy the underlying asset when its buy is unprofitable, but for this option, he has to bear a cost in the form of option premium charged due to time value of money and risk taken by the option writer.

Now, we can answer the question, **when should A use his right (technically known as exercising option)?** The decision to exercise a call option can be taken as follows:

Table 13.2: Decision to exercise a call option

Situations	**Decision**	**Market Terms**
Price of the underlying asset (S) at option expiry > Exercise price (E) i.e., $S - E > 0$	Exercise the call option because it is a profitable situation for the buyer of the call option.	In-the-money call option i.e., money will come to the option buyer.

Price of the underlying asset (S) at option expiry < Exercise price (E) i.e., S – E < 0	Don't exercise the call option because it is not a profitable situation.	Out-the-money call option i.e., money would flow out if the option is exercised.

Thus, a call option should be exercised when S > E or S – E > 0. The *exercise value or intrinsic value* (or just value denoted by VCO) of a call option at its expiry can be calculated using the following formula:

$$\mathbf{VCO} = \begin{cases} \mathbf{S - E} & \textbf{if } \mathbf{S > E} \\ \mathbf{0} & \textbf{if } \mathbf{S \le E} \end{cases}$$

A call option is exercised only when **S > E** because if a call option is exercised when **S < E,** then it will not be a rational decision because in that case receiving price S from the market by paying a higher amount E to the option writer is a loss-making transaction. Therefore, the value of the call option can also be expressed as this

VCO = Max [S – E, 0]

Max [S-E, 0] is read as chose the maximum of (S – E) and 0.

Profit on a call option is calculated using the following formula:

Net Payoff = VCO – option premium

If we plot the net payoffs against the prices of the underlying asset, then we get a graph known as the net payoff diagram. One point should be noted this diagram is drawn from **the option buyer's point of view**.

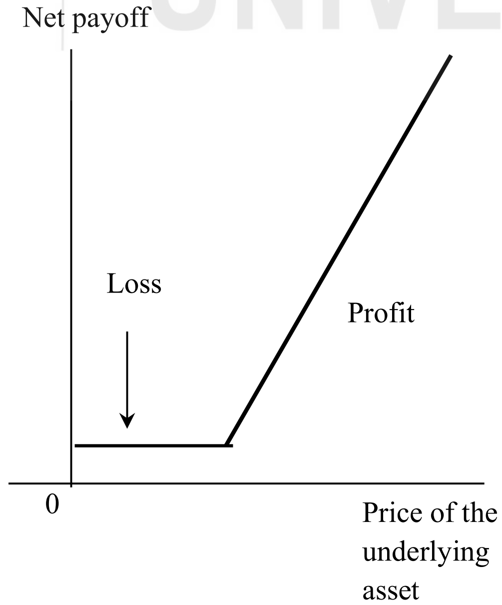


Fig. 13.1: Net payoff from the option buyer's point of view

If the net payoff diagram is drawn from the option writer's point of view, then the diagram is as follows:

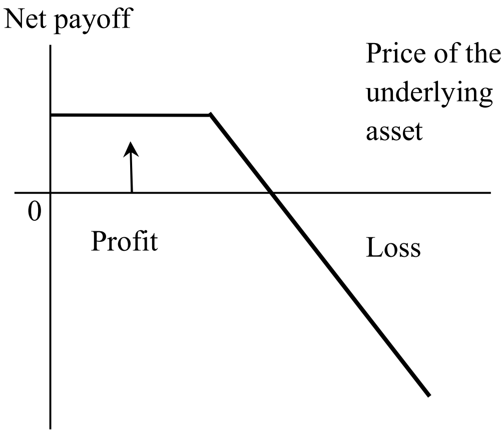


Fig. 13.2: Net payoff from the option writer's point of view

Let us understand Call option with the help of an example.

Example 13.2: Mr. Aakash is willing to buy a share of Reliance Industries and after 3 months, its probable prices are as follows: ₹ 5, ₹ 10, ₹ 15, ₹ 20, ₹ 25, ₹30. The current price is ₹ 7. Aakash purchases a call option from a derivative dealer and as per the terms and conditions, the exercise price is ₹ 12, and the option premium is ₹ 35.

a) You are required to suggest when to exercise the above call option.

b) Also, sketch the net payoff diagram from Aakash and the derivative dealer's point of view.

Solution

Net payoff call option buyer point of view i.e., Aakash.

Price of underlying asset (in ₹)	Option premium (in ₹)	Exercise price (in ₹)	Total cost (₹)	Net pay-off (₹)	Option exercise (Yes or No)
5	2	-	2	-2	No
10	2	-	2	-2	No
15	2	13	15	0	Yes
20	2	13	15	5	Yes
25	2	13	15	10	Yes
30	2	13	15	15	Yes

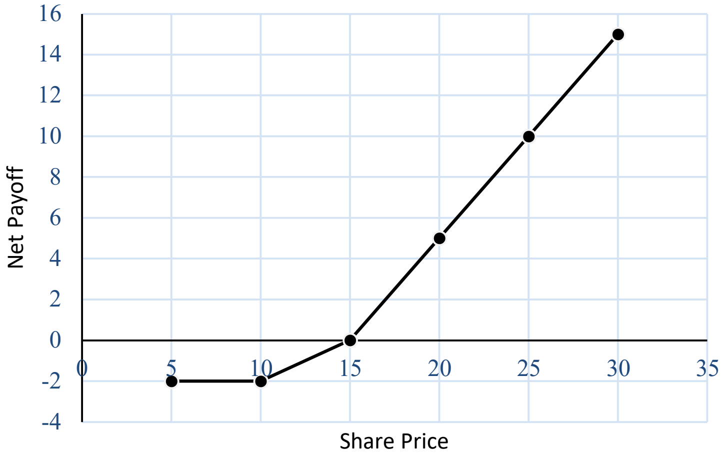


Fig. 13.3: Net Payoff Diagram from Akash's Point of View

Net payoff option writer point of view i.e., derivative dealer.

Price of underlying asset (in ₹)	**Option premium (in ₹)**	**Exercise price (in ₹)**	**Total Revenue (₹)**	**Net payoff (₹)**	**Option exercise (Yes or No)**
5	2	-	2	2	No
10	2	-	2	2	No
15	2	13	15	0	Yes
20	2	13	15	-5	Yes
25	2	13	15	-10	Yes
30	2	13	15	-15	Yes

Fig. 13.4: Net Payoff Diagram from option Writer's point of view

13.7.2 Put Option

A **put option** is a type of derivative in which the buyer gets a right to sell an underlying at an agreed price (known as exercise price or strike price) when the sale is profitable to him. Since this derivative is an option, therefore, he cannot be forced to sell the underlying asset when its sale is unprofitable, but for this option, he has to bear a cost in the form of an option premium. A put option can be exercised as follows:

Table 13.3: Decision to exercise a Put option

Situations	**Decision**	**Market Terms**
Exercise price (E) < Price of the underlying asset (S) at option expiry i.e., E– S < 0	Don't exercise the put option because it is a profitable situation for the buyer of the call option.	Out-the-money put option i.e., money will come to the option buyer.
Exercise price (E) > Price of the underlying asset (S) at option expiry i.e., E– S > 0	Exercise put option because it is not a profitable situation.	In-the-money put option i.e., money will come to the option buyer.

Thus, a put option should be exercised when E > S or E – S > 0. The *exercise value or intrinsic value* (or just value denoted by VPO) of a put option at its expiry can be calculated using the following formula:

$$VPO = \begin{cases} E - S & \text{if } E > S \\ 0 & \text{if } E \leq S \end{cases}$$

A put option is exercised only when E > S because if a put option is exercised when E < S, then it will not be a rational decision because receiving amount E from the option writer by purchasing for a higher amount S in the market is a loss-making transaction. Thus, the value of a put option can also be expressed as this

VPO = Max [E – S, 0]

Max [E-S, 0] is read as chose the maximum of (E – S) and 0.

Profit on a put option is calculated using the following formula:

Net Payoff = VPO – option premium

Example 13.3: Ms. Siddhi is willing to sell the share of SBI and after 2 months, its probable prices are as follows: ₹ 5, ₹ 10, ₹ 15, ₹ 20, ₹ 25, ₹30. The current price is ₹ 7. She purchases a put option from a derivative dealer and as per the terms and conditions, the exercise price is ₹ 12, and the option premium is ₹ 35.

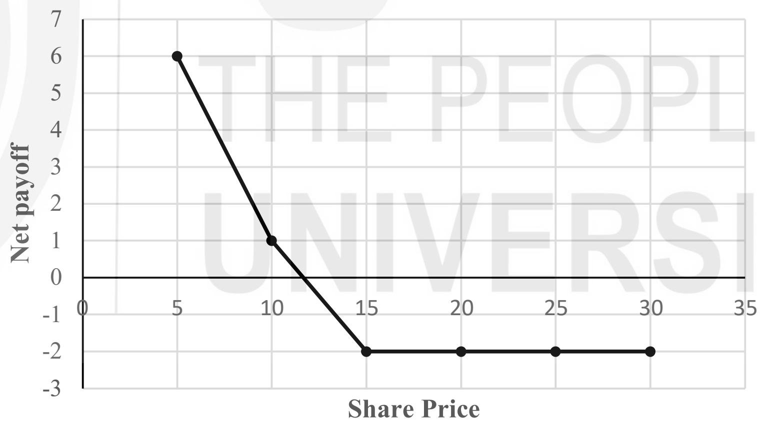
a) You are required to suggest when to exercise the above-put option.

b) Also, sketch the net payoff diagram from Siddhi and the derivative dealer's point of view.

Solution

Net payoff put option buyer point of view i.e., Ms Siddhi.

Price of underlying asset (in ₹)	Option premium (in ₹)	Exercise price (in ₹)	Total cost (₹)	Net pay-off (₹)	Option exercise (Yes or No)
5	2	13	7	6	Yes
10	2	13	12	1	Yes
15	2	-	2	-2	No
20	2	-	2	-2	No
25	2	-	2	-2	No
30	2	-	2	-2	No



Check Your Progress 2

1) What is an option? Why is it called an option?

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2) What are call and put options? Explain using an example.

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3) Do you think that option is a type of a future?

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4) When can the call option and put option be exercised?

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13.8 SWAPS

In finance, a swap refers to a financial contract between two parties to exchange cash flows or financial instruments. In this contract, the parties agree to swap one type of cash flow or financial instrument for another based on predetermined terms. These agreements encompass both spot and forward transactions within a customized framework. Corporations, banks, and individual investors now utilize swaps for intricate financing solutions, reducing borrowing costs, and gaining better control over interest rate and foreign currency risks. Swaps serve diverse purposes, ranging from hedging to speculation, but their core objective is to alter the nature of an asset or liability without the need for liquidation. For instance, an investor deriving returns from equity investments can transform those returns into less risky fixed-income cash flows without liquidating the equities. Similarly, a corporation with floating-rate debt can convert it into a fixed-rate obligation without retiring and reissuing debt.

Various features of Swaps are as follows:

1) **Customizability:** Swaps are highly customizable agreements tailored to meet the specific needs of the parties involved.

2) **Counterparty Risk:** Swaps involve counterparty risk since the agreement is dependent on both parties fulfilling their obligations.

3) **Over-the-Counter (OTC) Trading**: Swaps are typically traded OTC, which means they are not traded on standardized exchanges but rather through a network of dealers.

4) **Types of Swaps:** The most common types of swaps include interest rate swaps, currency swaps, and commodity swaps. Each type has different underlying assets and cash flow structures.

5) **Fixed and Floating Rates**: Swaps often involve the exchange of fixed interest rates for floating rates (or vice versa), helping parties manage interest rate exposure.

6) **Maturity:** Swaps have specified maturity dates, which can range from a few months to several years, depending on the agreement.

7) **Notional Principal**: The notional principal amount is the total value on which the exchanged cash flows are calculated. This principal itself is usually not exchanged.

8) **Cash Flow Exchange**: Parties exchange cash flows based on the agreed-upon terms, which may occur periodically (e.g., quarterly, semi-annually).

9) **Purpose:** Swaps are used for hedging risks, managing exposure to fluctuations in interest rates or currencies, and speculating on future price movements.

10) **Documentation**: Swaps are documented in a swap agreement, which includes terms such as the notional amount, payment dates, calculation methods, and conditions for default.

Let us understand various forms of swaps. Common swaps are:

a) **Interest rate swap –** Under this type of swap, one party agrees to pay fixed interest on a notional amount and receive floating interest on the same notional amount. Generally, the floating interest rate is calculated as LIBOR (London Inter Bank Offered Rate) + Some base points. In India, MIBOR (Mumbai Inter Bank Offered Rate) is also popular in place of LIBOR.

Example 13.5: Suppose there are two companies A and B. They enter an agreement with each other in which company A will pay Company B a fixed interest at 7% p.a. on a notional amount of ₹10,00,000 and it will receive floating interest i.e., LIBOR + 1%. The agreement is valid for 3 years. If the LIBOR data is given below, then the cash flows under the interest rate swap can be calculated as follows:

Years	**LIBOR**
0 to 1 year	8%
1 to 2 years	6%
2 to 3 years	10%

Step 1: Floating Interest

Years	**Floating interest rate = LIBOR + 1%**	**Floating interest (in ₹)**
1	9%	90,000
2	7%	70,000
3	11%	1,10,000

Step 2: Fixed Interest

Years	Floating interest rate = LIBOR + 1%	Floating interest (in ₹)
1	10%	100,000
2	10%	100,000
3	10%	100,000

Step 3: Exchange of net cash flows

Years	Net cash flows (in ₹)	Explanation
1	(-) 20,000	Company A will pay ₹ 100,000 and receive ₹ 80,000 from B, therefore, A can pay a net ₹ 20,000.
2	(-) 30,000	Company A will pay ₹ 1,00,000 and receive ₹ 70,000 from B, therefore, A can pay a net ₹ 30,000.
3	10,000	Company A will pay ₹ 1,00,000 and receive ₹ 1,10,000 from B, therefore, A can receive a net ₹ 10,000.

b) **Credit Default Swaps (CDS)** – A credit default swap (CDS) is a financial contract between two parties that provides insurance against the risk of default on a specific debt instrument, such as a bond or loan. The buyer of the CDS makes payments to the seller of the CDS in exchange for a guarantee that in the event of a default by the borrower, the seller will pay the buyer the face value of the debt instrument.

Example 13.6: Suppose Company A issues a bond worth ₹10 million with a maturity of 5 years. Investors who buy the bond are essentially lending money to Company A, and in return, Company A agrees to make regular interest payments and repay the principal at maturity. However, there is a risk that Company A may default on its debt obligations, which would mean that investors would not receive their interest payments or the principal at maturity. To mitigate this risk, an investor can purchase a CDS from a financial institution, such as a bank or hedge fund. The investor pays a premium to the institution to protect against the risk of default by Company A. If Company A defaults, the institution must pay the investor the face value of the bond, which is ₹10 million in this case. The investor effectively transfers the risk of default from themselves to the institution, which is compensated through the premium paid by the investor.

Check Your Progress 3:

1) State the key features of SWAP.

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2) Discuss the potential of SWAPS in India.

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13.9 LET US SUM UP

Derivatives: Financial contracts that derive their value from an underlying asset or security, such as stocks, bonds, or commodities constitute the derivatives. They have various pros and cons. Their biggest advantage is **hedging** against the risk and the biggest disadvantage is default risk. The various types of derivatives include Futures, Forwards, Options, Swaps etc.

Futures are Standardized contracts that obligate the buyer to purchase and the seller to sell a specific asset at a predetermined price and date in the future. Similar to futures contracts, **Forwards** are not standardized and are usually customized to meet the specific needs of the buyer and seller. **Options** are Contracts that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a predetermined price and date in the future. Agreements between two parties to exchange cash flows based on different financial instruments or indexes, such as interest rates or currencies are **Swaps**. There are various swap agreements, but interest rate swaps and credit default swaps are the two most popular ones. Under an interest rate swap agreement, two parties agree to exchange fixed interest for floating interest, while a credit default swap (CDS) is a financial contract that allows one party to transfer the credit risk of a particular asset or security to another party. In essence, it is a type of insurance policy that protects the buyer of the CDS from the risk of default by the issuer of the underlying security. Derivatives trading began in India in 2000, and the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) are the two exchanges in India where derivatives are traded.

Derivatives in India are traded on two main exchanges: the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE). The Securities and Exchange Board of India (SEBI) regulates the derivatives market. Commodity derivatives are traded on the Multi Commodity Exchange (MCX) and the National Commodity and Derivatives Exchange (NCDEX), while currency derivatives are traded on the Metropolitan Stock Exchange (MSE).

13.10 KEY WORDS

Derivative : A financial asset that derives its value from another asset called an underlying asset. There are various types of derivatives. It includes futures, forwards, options, commodity derivatives, etc.

Futures : Futures contracts are financial instruments that allow parties to buy or sell an asset, such as commodities, currencies, or financial instruments, at a predetermined price and date in the future. The price of the underlying asset is fixed at the time the futures contract is created, and the transaction is legally binding.

Forwards : Forwards are similar to futures, but they are not traded on exchanges. Instead, they are customized contracts between two parties to buy or sell an underlying asset at a predetermined price and time in the future.

Option : An option is a right to buy or sell an underlying at a strike or exercise price at the expiry of the option.

Swap : In finance, a swap is a derivative contract that allows two parties to exchange financial instruments or cash flows with each other based on agreed-upon terms. Interest rate swaps and credit default swaps are two very popular swaps.

13.11 SOME USEFUL BOOKS AND REFERENCES

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13.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A derivative is a financial instrument that derives its value from an underlying asset, such as a stock, bond, currency, or commodity. The value of a derivative is based on the changes in the value of the underlying asset.

2) Derives its value from an underlying asset, traded globally, Exchange-traded derivatives like futures or stock options are standardized and eliminate or reduce many of the risks of over-the-counter derivatives, usually leveraged instruments.

3) Lock in prices, hedge against risk, can be leveraged, useful in making a diversified portfolio. Limitations: difficult to value them, subject to counterparty default, complex, sensitive to demand and supply factors.

Check Your Progress 2

1) An option is an agreement between two parties (say A and B) in which a party, say, A, has the right to buy or sell a particular asset at a predetermined price within a specified time frame.

2) A **call option** is a type of derivative in which the buyer gets a right to **buy** an underlying at an agreed price. A **put option** is a type of derivative in which the buyer gets a right to sell an underlying at an agreed price.

3) No, Futures are a contract that obligates the buyer to purchase or sell an asset at a specified future date and price, while options give the buyer the right, but not the obligation, to purchase or sell an asset at a specified price and date.

4) Refer to Table 13.2 and 13.3

Check Your Progress 3

1) Refer to Section 13.8

2) Refer to Section 13.8